

# Modeling Normal Is All You Need: Joint Latent Clustering for Anomaly Detection in Multimodal Industrial-IoT Cyber-Physical Systems

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## Abstract

The Industrial Internet of Things (IIoT) instruments physical processes with networked sensors and controllers that stream multivariate telemetry, and monitoring this telemetry for faults and cyberattacks is essential to reliability and safety. Machine learning is the natural tool, but failures are rare and seldom labeled, so supervised detection is impractical: a detector must learn normal behavior from unlabeled telemetry alone. In cyber-physical systems this normal law is structured: a union of many imbalanced operating regimes shaped by physics and control, formalized as ten *Massive, Implicit, Imbalanced Multimodality* (MIIM) assumptions. That structure exposes a blind spot of the dominant approach: a fault can be individually normal on every sensor and reconstruct accurately, yet improbable under normal joint operation, so reconstruction-based detectors measure reachability rather than probability and miss it. Our detector, LatAD (**Latent-density Anomaly Detector**), learns a latent Gaussian-mixture clustering of regimes (VaDE, **Variational Deep Embedding**), scores anomalies by density, and factorizes that density over correlation-community subsystems combined by a cohesion-weighted **Higher Criticism** statistic. **On three real IIoT testbeds LatAD attains the best All-subset AUROC of any learned detector on HAI (0.948) and leads the difficult subset on HAI (0.845; a significant +0.088, 95% CI [0.042, 0.157]); USAD and TranAD fall to 0.44–0.50 on these reconstructable-but-improbable faults.**

**Keywords:** Industrial Internet of Things (IIoT); IoT anomaly detection; cyber-physical systems; industrial control systems; anomaly detection; unsupervised learning; sensor telemetry; fault detection

## 1 Introduction

A cyber-physical system (CPS) couples networked sensors, actuators, and controllers with a physical process, forming the sensing-and-actuation layer of the Industrial Internet of Things (IIoT); **monitoring its telemetry for faults and attacks is central to reliability and safety.** Such a system can be faulty even when every sensor reads within its normal range and a deep model reconstructs the full sensor vector accurately. In a tightly coupled plant, vehicle, or machine, normal behavior is spread across many operating regimes shaped by physics, actuator limits, and control logic, so a correlation break, or a state that falls

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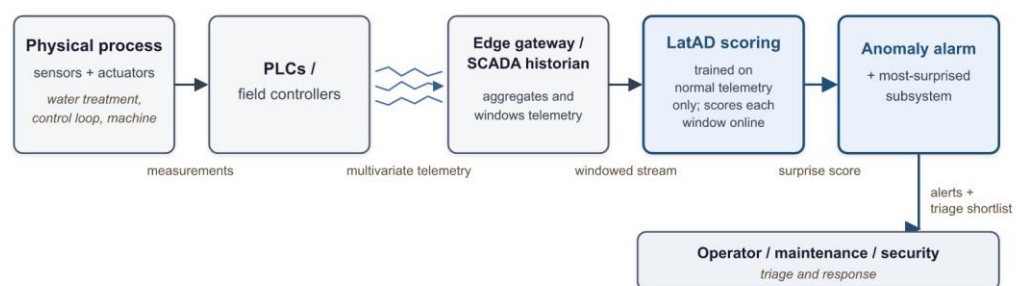
between two legitimate regimes, can be perfectly reconstructable yet deeply improbable. These are the faults that matter most for condition monitoring and predictive maintenance: early warnings that stay inside familiar sensor ranges while violating the joint structure of normal operation.

A modern CPS (a water-treatment plant, an industrial control loop, a rotating machine) integrates hundreds of **sensor and control** channels into a unit that continuously senses its own state and acts on it. This complexity puts its faults beyond enumeration: no test campaign can sample every combination of component, control regime, and operating condition in which a fault can arise, and many faults develop slowly, as wear or drift with no obvious symptom, until they surface as unplanned downtime or a safety-critical failure. Because the faults cannot be specified in advance, they must be discovered from the system's own behavior as it operates. This is the task of anomaly detection [1]: the **unsupervised or semi-supervised flagging of departures from a model of normal behavior**, adopted because labeled anomalies are too scarce to train a classifier [2, 3]. Normal telemetry accumulates continuously across a fleet of devices and operating regimes, whereas **each specific fault or attack is rare, diverse, and often system-specific** [4]. That behavior is richly observable: vibration, current, temperature, pressure, and position, together with internal control signals, stream continuously as IIoT telemetry and carry an early signature of almost any developing fault.

In an IIoT deployment this telemetry does not remain on the device: field sensors and programmable logic controllers feed edge gateways and Supervisory Control and Data Acquisition (SCADA) historians that aggregate, window, and forward the multivariate stream (Figure 1), so unsupervised anomaly detection is the operative safeguard for IIoT reliability and security [5]. The stakes are cyber-physical: an anomalous window can indicate sensor degradation, actuator malfunction, a developing process fault, or a control-plane manipulation or intrusion, and because the digital and physical layers are coupled, a missed anomaly can propagate from a tampered setpoint into the physical process [6].

### IIoT monitoring context: where LatAD sits

Multivariate telemetry is windowed at the edge, then scored online; LatAD flags anomalies and the most-surprised subsystem.



**Figure 1.** IIoT monitoring context. LatAD occupies the edge/historian tier: field sensors and programmable logic controllers (PLCs) feed an edge gateway or SCADA historian that aggregates and windows the multivariate telemetry, LatAD scores each completed window online, and an anomaly alarm together with the most-surprised subsystem is forwarded to the operator, maintenance, or security stack. Model fitting and calibration use train-normal telemetry only, so a site is commissioned from its own normal data without labeled attacks. Grey blocks are existing IIoT infrastructure; blue blocks are LatAD and its output.

Unsupervised anomaly detection for CPS telemetry is often framed as characterizing the anomalies, but that framing is misleading. With a representative set of labeled faults

the task would reduce to supervised classification; in practice genuine faults are rare and the few on record seldom span the ways a system can fail [3], so they cannot define the decision boundary, and they cannot even be used reliably for model selection. The usable information lies on the other side of the boundary, in the normal data, which is abundant and rarely exploited in full. Modeling that normal law faithfully is therefore the hard part of the problem. **Cyber-physical** normal is a law of a particular kind: shaped by deterministic physics, hard actuator and setpoint limits, and engineered control, it is the union of many bounded, curved operating regimes, some common and many rare, each with legitimately low-density fringes, rather than a single statistical blob. A detector that mis-models this structure fails however it thresholds.

A fault is anything that leaves this union of normal regimes, and the operationally difficult ones stay *inside* the marginal envelope of every channel while breaking the joint structure: a correlation break, a between-regime pocket, a history-inconsistent state. These faults are invisible to any per-channel range rule, and, as we show, **to the reconstruction residual that the deep detectors rely on, which on HAI places these faults no better than chance**, because the question is not whether a flexible decoder can reproduce the state but whether normal operation is likely to occupy it. Reconstruction measures reachability; detection needs probability.

Two obstacles have obscured this failure mode. The first is modeling. If normal is a union of many imbalanced regimes, a global anomaly model under-fits the regime structure, a single deep autoencoder alarms on rare-but-valid regimes (the trust-eroding false positive), **and the reconstruction residual that the deep detectors rely on misses exactly the reconstructable-but-improbable joint-structure faults that matter; where a fault does leave a reconstruction signal, a regime-conditioned residual stays competitive and is auto-gated back in.**

The second is evaluation. The dominant metric in the literature, point-adjusted **F1** (the harmonic mean of precision and recall), inflates scores so severely that a *random* anomaly score can beat every published deep model [7, 8]; and even under raw metrics, public CPS benchmarks are dominated by anomalies that a one-line univariate rule already separates [9], so a high headline number certifies little. Progress on the faults that matter cannot be measured until the easy majority is separated from the discriminative minority.

**A note on scope. Anomalies in a cyber-physical system arise from faults as much as from attacks: a drifting sensor, a worn actuator, or a slowly developing process fault alongside a deliberate intrusion. The operationally costly ones are frequently slow, an incipient malfunction that stays within the normal per-channel envelope while gradually crossing operating regimes, which is harder to catch than an abrupt, out-of-range event [10]. Open CPS datasets with real labeled anomalies are scarce and attack-centric: the few public testbeds inject discrete, largely abrupt attacks [9], while operational plant telemetry with labeled faults is seldom released for safety and security reasons. These testbeds are also small and single-mode (one nominal operating mode; §2.5). Slowly-progressing and between-regime faults are therefore underrepresented in public benchmarks, which shapes what any method can be exercised on and motivates the trajectory assumptions examined in §8.**

This paper makes three contributions. **(C1) A community-factorized latent-density detector, LatAD.** Its design follows an explicit list of ten structural assumptions of CPS normal data, MIIM (*Massive, Implicit, Imbalanced Multimodality*; §3), of which the reported model realizes seven (A1–A7): a jointly learned latent with explicit Gaussian-mixture regime clustering (Variational Deep Embedding, VaDE), scored in the latent by a mixture-density head and a rare-regime-safe nearest-component likelihood, factorized over the plant's own subsystems (communities of the normal-data correlation graph) and

combined by cohesion-weighted Higher Criticism (HC). This statistic flags when more subsystems are mildly surprised than chance allows, so a local fault that a single global density dilutes is concentrated and localized to its subsystem (§4), and thus each alarm carries an operator-facing subsystem-triage shortlist (evaluated on HAI and WADI, §8, Table 5). Reconstruction is demoted and re-enters only through a gated residual head.

**(C2) A detector-agnostic difficulty stratification.** A channel-based rule separates the trivially detectable anomalies and a linear principal-component subspace filter (Hotelling  $T^2$  and squared prediction error) separates the linearly detectable ones, leaving a difficult and a double-hard subset that the detector under test plays no part in constructing (§5–6).

**(C3) A drift-aware anomaly typing.** A causal per-community decomposition into a slow on-trend baseline and a fast residual types each window as gradual drift or abrupt change-point; scoring the changepoint component recovers detection on the severely non-stationary SWaT record (difficult-subset area under the ROC curve (AUROC), from 0.524 to 0.723) and cuts its normal-window false-alarm rate from 79 to 6.5 percent, while leaving stationary records unchanged (§7). We evaluate the deep detectors USAD, TranAD, and GDN under the same raw point-wise metrics. LatAD leads the difficult subset on HAI, significantly, where those detectors fall to near chance, and leads every evaluated nonlinear detector on WADI, statistically tied with the linear baseline; because aggregate benchmark scores are dominated by easy marginal excursions, the difficult subset, not the headline number, is the central test throughout.

The study covers three datasets with a single trained configuration per dataset; AUROC is reported as a five-seed mean, and the best-F1 threshold is a test-swept oracle disclosed in §5.

## 2 Related work

On time series, anomaly detection must respect temporal and cross-channel dependence [11], and on cyber-physical and industrial control systems it underpins physics-aware intrusion and fault detection [12].

### 2.1 Anomaly detection in the Industrial IoT and edge intelligence

Anomaly and intrusion detection is a core reliability-and-security function of the Industrial Internet of Things, where networked sensing, actuation, and industrial processes are monitored continuously from streaming telemetry rather than from packet-level network traces alone [5, 6]. The field leans on unsupervised, one-class, and normal-only models: deep autoencoders and related density models are trained on normal IoT telemetry and flag departures from it [4, 13], a setting LatAD shares. A parallel thread pushes detection to the network edge: moving inference onto gateways and industrial edge nodes reduces telemetry backhaul and enables local response, at the cost of compute, memory, and latency budgets that shape model design [14, 15, 16]. IIoT deployments are also heterogeneous and multimodal at fleet scale, with many devices, subsystems, and operating regimes whose distributional structure a single global model captures poorly [6]. Global autoencoders trained across such heterogeneity develop detection blind spots on multi-product CPS that grade-specific models recover [17], reinforcing our factorization of the density over correlation-community subsystems rather than a single plant-wide model. Process-level, subsystem-localized detection from normal-only telemetry, under regime imbalance, remains open.

### 2.2 The illusion of progress in time-series anomaly-detection evaluation

A line of critical work shows that the standard evaluation of time-series anomaly detection is unreliable. Kim et al. [7] demonstrate that the ubiquitous *point-adjustment* protocol (marking an entire ground-truth anomaly segment as detected if any single point in it is flagged) inflates F1 so drastically that a *random* anomaly score achieves higher point-

adjusted F1 than every state-of-the-art model on SWaT and WADI. Garg et al. [8] independently reproduce this inflation and show that simple baselines (principal-component analysis, PCA, and channel-wise autoencoders) beat elaborate deep models once metrics are made rigorous, and that deep methods often fail to detect even simple anomalies. Doshi et al. [18] diagnose the point-adjustment reward structure and propose a rectified evaluation, and proximity- and event-aware metrics such as PATE [19] move beyond point adjustment altogether. The critique now spans metrics and benchmarks alike. Wu and Keogh [9] argue that many popular benchmarks are flawed (triviality, mislabeling and unrealistic anomaly density) creating an illusion of progress; and a recent benchmark analysis [20] finds that anomaly segments in multivariate CPS benchmarks are mostly univariate (SWaT and WADI sit among the datasets whose anomalies deviate univariately on close to all timesteps, with no long cross-channel-only segments), so a flat univariate model matches channel-dependent deep detectors. Recent large-scale benchmarks and unified benchmark suites confirm that detector rankings are strongly dataset-dependent [21, 22], which is why we read every method per dataset and per difficulty subset rather than by a single headline number.

A parallel critique targets the dominant scoring mechanism itself. Reconstruction-based detectors assume that a model trained to reconstruct normal data reconstructs anomalies worse; Bouman and Heskes [23] show this assumption is unreliable, because anomalies lying far from normal data can be reconstructed perfectly in practice, and Gong et al. [24] earlier showed that a deep autoencoder generalizes well enough to rebuild anomalies and miss them. This motivates scoring in a clustered latent rather than through a reconstruction residual. A position paper by Sarfraz et al. [25] draws the two threads together, characterizing the field as plagued by flawed evaluation metrics, inconsistent benchmarking, and unjustified design choices, and showing that state-of-the-art deep detectors effectively learn linear mappings that simple baselines reproduce; Wagner et al. [26] re-audit the widely used multivariate benchmarks, discard the ones with erroneous labels, and note the absence of any standard evaluation metric.

### 2.3 Latent and clustering models for anomaly detection

Latent and clustering models detect anomalies by learning a representation of normal and scoring departures in that space. Variational Deep Embedding (VaDE) [27] pairs a variational autoencoder with a Gaussian-mixture latent prior, learning representation and clusters jointly, and DAGMM [28] couples an autoencoder with a Gaussian-mixture energy. One-class methods place normal inside a learned boundary: Deep SVDD [29] maps normal points near a fixed latent center and scores by distance to it, and THOC [30] extends this to multiscale temporal one-class hyperspheres; OmniAnomaly [31] scores multivariate telemetry by reconstruction probability under a stochastic-recurrent latent. These methods all score against a single global model of normal; none estimates density per physical subsystem under regime imbalance, the gap this paper addresses. Closest to our aim, Birihanu and Lendák [32] flag industrial-control-system (ICS) anomalies from a latent correlation matrix scored by a multivariate Gaussian, whereas we factorize a learned latent density over correlation communities and combine them by Higher Criticism. Standard tools used later are Ledoit–Wolf shrinkage [33] for whitened residuals, and Isolation Forest [34] and deep autoencoders [35] as baselines.

### 2.4 Deep multivariate CPS anomaly detection

USAD [36] couples two adversarially trained autoencoders; TranAD [37] uses deep transformer networks with adversarial and self-conditioning training; GDN [38] learns a graph over channels and detects deviations from learned inter-channel relations. Graph mixture-of-experts routing likewise injects hierarchical graph structure into entity representations for GNN detectors [39], whereas we route per-community density experts by cohesion-

weighted Higher Criticism rather than a learned gate. These define a widely used deep state-of-the-art (SOTA) on SWaT/WADI-style benchmarks. More recent detectors refine rather than abandon reconstruction: SensitiveHUE [40] models per-variable heteroscedastic uncertainty to sharpen sensitivity to normal patterns, and CATCH [41] learns channel correlations through frequency-domain patching. Both remain reconstruction-based, so the mechanism analysis of §8 suggests they inherit the same blind spot on the reconstructable-but-improbable joint faults that define our difficult subset. We benchmark USAD and TranAD, the reconstruction- and transformer-based deep detectors, under the same raw, difficulty-stratified metrics (the comparison is in §6). We also benchmark GDN under the same protocol (§5.5). A raw-metric evaluation of the newer reconstruction detectors (SensitiveHUE, CATCH) is left to future work.

Beyond public benchmarks, sensor-driven condition monitoring and fault prediction in operational CPS motivate modeling the multivariate sensor stream directly: vibration-based machine diagnostics [42, 43], structural-health monitoring of rotorcraft components [44], and fault prediction in rotorcraft flight controls [45].

## 2.5 Benchmark versus operational scale

Per-dataset details are given in §5.1. The water-treatment and ICS-security benchmarks used here (SWaT, WADI, HAI) and the related BATADAL [46] share a scale profile: roughly 40 to 120 channels, days to weeks of data, and a single nominal operating mode. This is far below operational scale. BATADAL extends to about a year but keeps only 43 hourly variables, and N-CMAPSS [47] reaches a 40-unit fleet with full run-to-failure trajectories but exposes only 14 sensors; operational sources such as the Building Data Genome Project 2 [48] (3,053 meters across 1,636 buildings over two years) and plant process historians carrying thousands to tens of thousands of tags span orders of magnitude more channels, far longer horizons, and many operating modes. A parallel line of work argues that benchmark realism, not model novelty, now limits measured progress [9, 25, 22]. Operational scale is therefore the setting in which the trajectory assumptions A9 and A10 and the masking form of A8 become testable.

## 3 The MIIM assumptions about IIoT/CPS normal telemetry

We model the normal law of a CPS as a mixture over reachable operating regimes,

$$p(x) = \sum_{k=1}^K \pi_k p_k(x) \quad (1)$$

where each component  $p_k$  is a bounded, oriented, curved patch of observation space. The ten assumptions of Table 1 describe the qualitative structure of this mixture; each is grounded in a physical property of CPS operation and A1–A7 each motivate a specific design choice in the detector of §4. These properties are not a universal law but a checklist that CPS normal may exhibit to varying degrees: a given dataset realizes some strongly, others weakly or not at all. LatAD is built to exploit each of A1–A7 when it is present and to reduce to a standard latent-density model when it is absent. We call the resulting property MIIM: the sample is Massive, the regime structure is Implicit (unlabeled and only approximately recoverable), the regime occupancy is Imbalanced, and normal is Multimodal. Two consequences matter for detection. Normality is multimodal in regime space; and because the few levers of A4 drive correlated channel groups (A7), deviations are often local to one tightly coupled subsystem. LatAD therefore models regime probability at subsystem scale, not only at whole-plant scale (§4.4).

**Table 1.** The ten CPS normal-data assumptions and their physical origin. A1–A8 shape the instantaneous window; A9–A10 shape the trajectory. The reported model realizes A1–A7; A8–A10 are specified but not realized by the window-only model. Each realized assumption is mapped to the

LatAD design section that exploits it in §4, and the realized-versus-future scoping is discussed in §8.

| ID  | Assumption                                                         | CPS rationale                                                                                                                                                                                                                                                                                                                                                 |
|-----|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1  | <b>Regime mixture</b> (multimodality)                              | A plant or vehicle cycles through discrete operating regimes (idle, cruise, load steps); normal is their union, not one blob.                                                                                                                                                                                                                                 |
| A2  | <b>Regime explosion</b> (exponential regime growth)                | A CPS is built from many interacting subsystems; each added element multiplies reachable regimes, so regime count grows with complexity.                                                                                                                                                                                                                      |
| A3  | <b>Thin fringes</b> (intra-regime sparsity)                        | Within a regime the system dwells near a typical operating point and reaches the edges only occasionally; each regime holds legitimate low-density regions near its boundary.                                                                                                                                                                                 |
| A4  | <b>Few levers</b> (low intrinsic dimension)                        | The independent control variables are far fewer than the observed channels; the rest follow deterministically or probabilistically.                                                                                                                                                                                                                           |
| A5  | <b>Heavy tail</b> (Zipf imbalance)                                 | Machines spend most of the record in a few steady regimes; startup, shutdown and rare maneuvers occupy little of it, so an infrequent-but-normal point is easily mistaken for a fault.                                                                                                                                                                        |
| A6  | <b>Hidden regimes</b> (latent, implicit regimes)                   | True regimes are an emergent artifact of coupling and are not logged; regimes must be discovered approximately from raw streams.                                                                                                                                                                                                                              |
| A7  | <b>Mixed signals</b> (heterogeneous scale and noise)               | The bus carries sensors, actuators, discrete states and setpoints, each with its own range, resolution and correlated noise.                                                                                                                                                                                                                                  |
| A8  | <b>Between-regime overlap</b> (heavy mode overlap between regimes) | Neighboring regimes could overlap in observation space, so a window would fall in a zone of mode ambiguity while every channel stays in range. The precondition (adjacent-but-distinct regimes) holds on nearly every plant, but heavy mode overlap is absent under valid observation-space metrics, and the masking form is coupled to A10 (§8, Appendix C). |
| A9  | <b>Many clocks</b> (multiscale dynamics)                           | Physical inertia sets per-channel time constants (thermal slow, electrical fast); only some regime→regime transitions are legal, each with a characteristic dwell.                                                                                                                                                                                            |
| A10 | <b>Path dependence</b> (history-conditioned normality)             | A fault is defined relative to operational history: the same instantaneous state can be normal or faulty depending on how the system arrived there.                                                                                                                                                                                                           |

Two standing qualifiers apply throughout: the sample is **massive** ( $N$  large) and the mixture is approximately **stationary** within a run. Measured on train-normal, the MIIM structure is present but *implicit* (A6): cluster sizes are heavy-tailed, and Bayesian-information-criterion (BIC) regime counts keep improving past two dozen components (BIC-preferred  $K^*$  of 22, 24, and 25 on WADI, HAI, and SWaT, on a component grid extended to 25; the SWaT value sits at the grid ceiling and is therefore a lower bound). The regimes are **least separated** on WADI and HAI (silhouette 0.06 and 0.08, on a scale where 1 is well-separated clusters and 0 is **adjacent, touching clusters**) but separate more crisply on SWaT (0.29). This ordering tracks where the design pays off: **the regime-community factorization gains most on WADI, the least-separated case, and by smaller margins on HAI and SWaT, since a single global density blurs local structure (§6).**

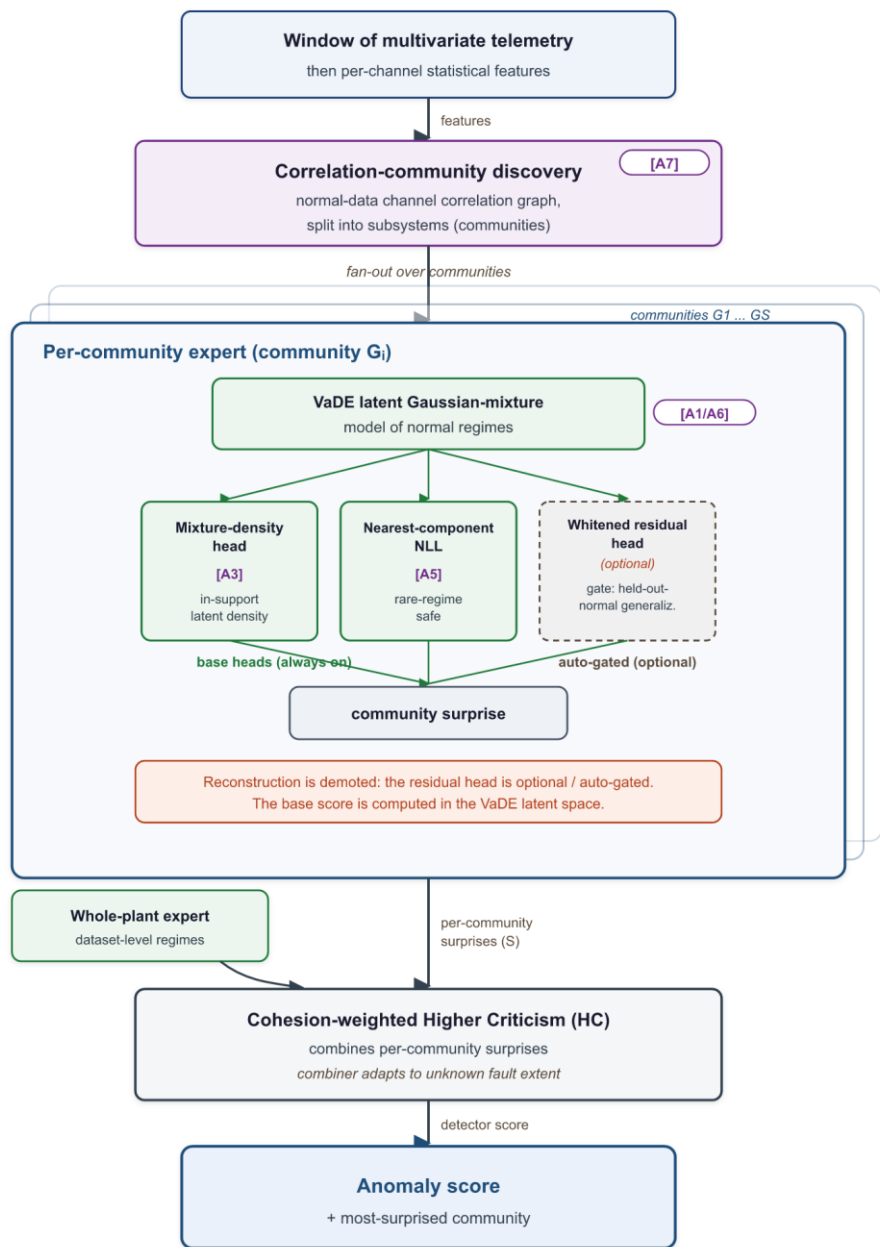
## 4 Method

We call the detector **LatAD**, and preview its pipeline (Figure 2) before the details: windowed multivariate IIoT telemetry → correlation-community subsystems → one regime-density model per community plus a global expert → calibrated per-community tail probabilities → cohesion-weighted Higher Criticism → anomaly score. Two stages implement it. A representation stage, a VaDE, jointly learns a latent and a Gaussian-mixture regime prior; a scoring stage computes a stack of density heads in that latent, all calibrated on train-normal. The reported detector, **LatAD (regime-community)**, estimates this density per subsystem (§4.4); the base ablation **LatAD (global density)** estimates it once over the

whole channel set. Because every community uses the same representation and heads, we describe those first, then the regime-community realization.

LatAD workflow

Subsystem-aware latent detector: per-community VaDE experts, latent surprise heads, and an HC combiner.



Tags [A#] mark the MIIM assumption each block realizes:  
[A7] correlated channels / subsystems · [A1/A6] jointly learned regimes · [A3] mixture density · [A5] rare-regime safe · [A7] auto-gated whitened residual

Figure 2. LatAD workflow. A window of multivariate telemetry is reduced to per-channel statistical features; the normal-data channel correlation graph is split into subsystems (communities, A7); and each community, together with a whole-plant expert, is scored by a VaDE latent Gaussian-mixture model of normal regimes (A1/A6) through two always-on base latent heads (mixture density, A3;

nearest-component NLL, A5) and one optional auto-gated whitened-residual head (A7). Reconstruction is demoted and the base score is computed in the latent space. The per-community surprises are combined by cohesion-weighted Higher Criticism into the anomaly score and the most-surprised community. Bracketed tags mark the MIIM assumption each block realizes.

The reported detector realizes A1–A7 (each mapped to a design section below); A8–A10 are specified and measured but not realized (§8).

**One realization of the assumptions.** A1–A10 specify *what* a model of CPS normal must capture, not *how* to implement it; the detector below is one concrete realization of A1–A7, reported in full so that every choice is auditable, and §8 surveys the design space the same assumptions admit. Each realized assumption maps to a specific mechanism: A1 (regime mixture) and A6 (latent regimes) to the jointly learned VaDE Gaussian-mixture latent (§4.1); A2 (regime explosion) to the high component count and mixture-density head (§4.1, §4.3); A3 (thin fringes) to the mixture-density head (§4.3 i); A4 (low intrinsic dimension) to the low-dimensional latent (§4.1); A5 (heavy-tailed occupancy) to the nearest-component likelihood (§4.3 ii); and A7 (heterogeneous scale) to per-feature standardization with a whitened residual (§4.3 iii) and the community factorization (§4.4).

#### 4.1 Representation: Variational Deep Embedding (VaDE) (the representation stage; Figure 2, top)

VaDE [27] is a variational autoencoder (VAE) whose latent prior is a Gaussian mixture, so the representation and the operating-regime clusters are learned *together* (A1, A6) rather than one after the other. The encoder and decoder widths, mixture size, training schedule, and scoring-head settings are listed in Appendix D (Table D1). An encoder maps a window feature vector  $x$  to a latent Gaussian  $q(z | x) = N(\mu, \text{diag } \sigma^2)$ ; a decoder reconstructs  $\hat{x}$ .

The latent prior is the Gaussian mixture,

$$p(z) = \sum_{c=1}^K \pi_c N(z | \mu_c, \text{diag } \sigma_c^2), \quad (2)$$

with mixture parameters  $\{\pi_c, \mu_c, \sigma_c\}$  learned jointly with the networks. The training objective is the negative evidence lower bound; per window it is

$$\mathcal{L} = \sum_j (x_j - \hat{x}_j)^2 + \beta \sum_c \gamma_c \left[ \text{KL}(q || N(\mu_c, \sigma_c^2)) + \log \frac{\gamma_c}{\pi_c} \right], \quad (3)$$

where  $\gamma_c \propto \pi_c N(z | \mu_c, \sigma_c^2)$  are the responsibilities (the posterior probability that a window belongs to regime  $c$ ); the reconstruction term sums over the 6C window features, indexed by  $j$ . Three standard measures prevent the well-known cluster collapse in which many components are abandoned during joint training: a Kullback–Leibler (KL)/cluster *warm-up* that anneals  $\beta$  from 0 to 1 so the encoder settles on the pretrained mixture initialization before the prior is pulled around; a DAGMM-style [28] penalty on tiny component variances (a variance floor plus a  $\sum_c \sigma_c^{-2}$  term); and a slower learning rate on the mixture parameters so the Gaussian mixture model (GMM) initialization (fitted on the pretrained latent) is refined, not destroyed. Training proceeds as plain-VAE pretraining, GMM initialization on the encoded means, then joint optimization of the full objective.

#### 4.2 Scoring: density over reconstruction

The natural VaDE anomaly score is the joint negative log-likelihood (NLL): a whitened reconstruction residual plus a latent term. On the *difficult* faults (correlation breaks and between-regime pockets) this summed score can be counterproductive. On the HAI difficult subset the naive joint negative log-likelihood falls below the latent term alone, so the whitened reconstruction residual, informative in isolation, drags the sum down; the reported model demotes reconstruction to a gated head (§4.3 iii, Table A5).

The mechanism is that a flexible decoder can *reconstruct a fault faithfully*: a correlation-break window has in-range marginals, so the decoder rebuilds it and the residual carries little signal. This is what happens on HAI, where adding the whitened residual pulls the summed joint score below the latent term alone, so demoting reconstruction helps. The base scoring stack therefore works entirely in the jointly learned latent; reconstruction re-enters only through the optional, auto-gated residual head, which fires only where it generalizes to held-out normal (§4.3 iii).

#### 4.3 The scoring stack (the scoring stage; Figure 2, per-community heads)

Let  $z = \mu(x)$  be the encoder mean. The base score sums two latent heads, each **standardized (recentered and rescaled)** against its train-normal mean and standard deviation, so the operating scale is set by normal, not by the test batch.

(i) Latent mixture-density head (a high-resolution Gaussian-mixture density estimator for non-Gaussian pockets)

Because each regime has a non-uniform interior with thin fringes (A3), one Gaussian per regime is too coarse. We fit a high-resolution diagonal Gaussian mixture on the train-normal latent ( $M = 80$  components by default, distinct from the VaDE regime count  $K$ ) as a high-resolution Gaussian-mixture density estimator, and score a window by its density negative log-likelihood:

$$s_{\text{dens}}(x) = -\log \sum_{m=1}^{80} w_m N(z \mid a_m, \text{diag } b_m^2). \quad (4)$$

Here  $w_m$ ,  $a_m$ , and  $b_m^2$  are the weight, mean, and diagonal variance of the  $m$ -th of the 80 density-mixture components (a separate high-resolution mixture fit on the train-normal latent, distinct from the VaDE regime mixture).

(ii) Nearest-component NLL (rare-regime-safe)

Under heavy-tailed imbalance (A5) a valid point in a rare, low- $\pi$  regime must not be flagged solely for being rare (Appendix E demonstrates this rare-regime safety on a dataset with imbalanced regime occupancy). We therefore use the *closest* component, not the  $\pi$ -weighted mixture:

$$s_{\text{near}}(x) = -\max_c \log N(z \mid \mu_c, \text{diag } \sigma_c^2). \quad (5)$$

Using the maximum over components rather than the Bayesian mixture sum keeps a rare-regime signal that the mixture sum washes out (A5): the nearest-component distance retains a deviation confined to a single sparsely populated regime, whereas the mixture sum averages it away. This term is a rare-regime safety head: its role is to keep a valid window in a sparsely populated regime from being flagged for rarity alone (A5), not to maximize difficult-subset AUROC. On a dataset without such rare-regime false alarms it can leave the base score marginally below the density head alone (Table A5, WADI). On these three benchmarks its rare-regime false-alarm rate is not measurably below the mixture head's; the safety gain is demonstrated on the imbalanced 3W record (Appendix E, Table E2). The base score is the sum of the two **standardized** heads,  $s_0 = \text{std}(s_{\text{dens}}) + \text{std}(s_{\text{near}})$ .

(iii) Optional responsibility-weighted whitened residual, auto-gated

On some datasets the fault *does* surface in reconstruction. The optional residual head reduces the per-channel residual  $r$  by principal-component analysis (fit on train-normal) to 30 dimensions, fits a Ledoit–Wolf-shrunk [33] (a covariance estimate regularized toward the identity, stable when channels outnumber samples) covariance  $\Sigma_k$  per regime, and combines the per-regime whitened energies of that reduced residual by responsibility:

$$s_{\text{resid}}(x) = \sum_{k=1}^K \gamma_k r_k^\top \Sigma_k^{-1} r_k. \quad (6)$$

Here  $r_k$  is the PCA-reduced residual and the quadratic form is its whitened (Mahalanobis) energy under regime  $k$ 's covariance  $\Sigma_k$ , and  $\gamma_k$  is the window's responsibility for regime  $k$ . The head is *auto-gated* by a held-out-normal generalization test: per-regime precisions are fitted on the first 80% of train-normal (split A) and scored on the last 20% (split B); if held-out normal scores much higher (the residual overfits or drifts), the ratio  $q_{95}(B)/q_{95}(A)$  exceeds 1.5 and the head is switched off. Empirically the ratio is 3.41 on WADI (off), 1.22 on HAI (on), and 0.76 on SWaT (on): the residual generalizes to held-out normal on HAI and SWaT but not on WADI, and the gate recovers this pattern from train-normal alone.

#### (iv) The combined window score

The reported detector scores a window by summing the base score and, where the residual head is gated on, the whitened residual:

$$s(x) = s_0 + g s_{\text{resid}}. \quad (7)$$

Equation (7) is the detector's complete window score, with  $g \in \{0,1\}$  the train-normal auto-gate on the residual head (§4.3 iii;  $g = 0$  on WADI). The reported model configures its heads purely from train-normal statistics, with no test-set tuning.

#### 4.4 Regime-community density (the headline realization and subsystem-factorization stage; Figure 2, community fan-out and HC combiner)

The stack of §4.1–4.3 estimates one density over the whole channel set. Its headline realization estimates density *per subsystem* instead, and the reason is physical. A CPS is an assembly of coupled subsystems, process units, control loops, and conservation-linked flows, each binding a small group of sensors and actuators through tight local correlation while remaining only loosely coupled to the rest of the plant (WADI's parallel treatment trains, SWaT's six sequential stages, HAI's coupled boiler, turbine, and water processes). Mass and energy balances correlate the channels within a unit, a control loop ties a measurement to its setpoint, and a shared control lever drives its dependent channels together, so the few independent levers of A4 and the heterogeneous, correlated channels of A7 surface as blocks of the channel correlation graph. Agglomerative clustering on the train-normal correlation graph groups the channels into nested communities  $G$  (dendrogram subtrees of size 3 to 25 channels) that serve as data-driven proxies for these subsystems, with no supervision and no plant schematic.

Faults inherit the same locality: a stuck valve, a drifting analyzer, or a fouled sensor perturbs its own subsystem while the rest of the plant runs normally, so an incipient fault is typically confined to one community, a handful of channels out of hundreds. A single global density dilutes such a local violation across the full dimensionality (a single-group shift is a vanishing fraction of WADI's 732-dimensional feature, 122 channels  $\times$  6 statistics), whereas a per-community density concentrates it. For each community  $G$  we therefore fit the full detector of §4.1–4.3 on that community's channels alone (a separate small VaDE per community, each with its own encoder, Gaussian mixture, and scoring heads, its regime count and latent dimension scaled to the community size  $|G|$ ; nothing is shared across communities except the input window features, so the model is a stack of independent per-subsystem experts, tens per plant, plus one global expert). We read its calibrated surprise  $s_G$  (the train-normal upper-tail negative log-probability), and we include the global detector as an unfactorized *whole-plant* expert so the detector retains sensitivity to dense whole-system faults. Summing the per-community surprises gives a simple pooled score (because the communities are nested subtrees and can overlap, this sum is an over-complete pooling rather than an exact factorization),

$$s_{\text{pool}}(x) = \sum_{G \in \mathcal{C}} s_G(x), \quad (8)$$

and the detector scores a window by a weighted combination  $\bigoplus_{G \in \mathcal{C}} w_G s_G(x)$ . The weight  $w_G$  grows with the community's internal cohesion (its mean absolute pairwise channel correlation), because a density violation inside a tightly coupled subsystem (A7) is more diagnostic than the same deviation in a loosely coupled one. The combiner  $\bigoplus$  must adapt to the fault's unknown extent: summation is optimal when the fault perturbs many communities, a maximum when it is confined to one, and Higher Criticism [49], a sparse-signal statistic that asks whether more communities are mildly surprised than chance allows, interpolates between the two without assuming the extent in advance. Writing the per-community upper-tail p-values for the  $S$  community experts in ascending order  $p_{(1)} \leq \dots \leq p_{(S)}$ , the statistic is

$$HC = \max_{i: p_{(i)} < 1/2} \frac{\sqrt{S} (i/S - p_{(i)})}{\sqrt{p_{(i)}(1-p_{(i)})}} \quad (9)$$

over the surprising tail. Cohesion weighting raises each p-value to the power  $w_G/\bar{w}$  before sorting, with  $w_G = \text{coh}_G \sqrt{|G|}$  (community cohesion times root size) and  $\bar{w}$  their mean, so a violation in a tight, large subsystem weighs more; every p-value is a train-normal upper-tail probability, so the combiner is calibrated on normal alone. The factorization also localizes: the per-community surprises rank the plant's subsystems by how implicated they are, giving a subsystem triage shortlist (§8, Table 5). The reported **LatAD (regime-community)** combines the communities by cohesion-weighted Higher Criticism and fuses that with the **whole-plant** expert by summing the two in train-normal standardized units; the single-latent **LatAD (global density)** of §4.1–4.3 is the degenerate case  $\mathcal{C} = \{\text{all channels}\}$  and is reported as the base ablation. The detector occupies one point on each of four design axes: representation (VaDE), density estimation (itself two decisions, community construction and surprise aggregation), scoring heads, and calibration. Appendix A ablates the density-estimation axis.

## 5 Experimental methodology

### 5.1 Datasets

We use three real CPS datasets and no synthetic data in the main results tables (Tables 2 to 5). All three are instrumented Industrial-IoT / industrial-control-system testbeds: networked sensors, actuators, and programmable logic controllers stream multivariate telemetry from a live physical process, and each carries real executed attacks, making them standard proxies for IIoT security-monitoring and condition-monitoring workloads.

- **WADI** [50], a water-distribution testbed (the WADI.A1 April-2017 release), 123 sensor/actuator channels. Roughly a quarter of channels are near-constant in normal operation; these are attack *carriers*, not noise, and are kept. One channel, 2B\_AIT\_002\_PV, is rescaled between the normal and attack recordings and is dropped; WADI results use the remaining 122 channels. WADI is downsampled 10× on both streams.
- **HAI** [51], a hardware-in-the-loop (HIL, a real controller driving a simulated plant) industrial control system testbed (HAI 20.07), 59 channels, a multi-process plant. No clipping (its large excursions are real attack signal).
- **SWaT** [52], the Secure Water Treatment testbed from iTrust, a 51-channel six-stage water-treatment plant with real executed attacks. Standardized features are clipped to  $\pm 10\sigma$  on SWaT and WADI so recalibrated analyzer channels (the SWaT AIT201 channel is a documented example [53]) do not dominate the per-channel range statistics; HAI is not clipped. SWaT is downsampled 10× on both streams. We use the December 2015 iTrust SWaT release: the SWaT normal recording supplies train-normal (its first 2%, the plant start-up and stabilization transient, is

excluded before training), and the SWaT attack recording (449,919 one-second rows over 51 channels, 12.14% of timesteps labeled attack) supplies the test set, whose normal periods interleave between attack episodes.

Temporal 10× downsampling of WADI and SWaT (both natively 1 Hz over many days) is standard practice on these benchmarks: it matches the 10-second median protocol of GDN [38] and lies within the downsampling range USAD [36] reports as leaving detection performance largely unchanged, with the train and test horizons matched on both testbeds. HAI is used at native resolution.

## 5.2 Windows and features

Each raw channel is standardized to zero mean and unit variance using train-normal statistics only (A7). We then slide a fixed window of length  $W = 60$  with stride 30 across the standardized stream. Each window is summarized by six per-channel statistics: mean (level), standard deviation (variability), minimum, maximum, first-to-last difference (net trend), and range, concatenated into one flat feature vector, so a  $C$ -channel window becomes a  $6C$ -dimensional input. A window is labeled anomalous if more than 5% of its timesteps are attack timesteps. Model inputs are additionally per-feature standardized on train-normal before the encoder. The window size and stride are unified across all three datasets. The reported model uses these static per-channel features. A controlled representation check (Appendix A, Table A2) confirms that the latent-density mechanism, not the choice of window statistics, carries the WADI result; on HAI and SWaT a richer temporal representation is a complementary lever (+0.071 and +0.241 difficult-subset AUROC, Appendix A, Table A2), aligned with A9 and left to future work (§8).

**Model configuration.** The VaDE uses  $K$  mixture components and latent dimension set per dataset (WADI 20 and 10, HAI 40 and 16, SWaT 40 and 16); the density head is a diagonal mixture of 80 components (capped at one tenth of the train-window count), and the residual head is reduced to 30 dimensions. Correlation communities are agglomerative-clustering (average-linkage) subtrees of size 3 to 25 on the train-normal channel-correlation graph, yielding 44, 28, and 24 communities on WADI, HAI, and SWaT. All settings are shared across datasets except the per-dataset mixture size and latent dimension.

## 5.3 Difficulty stratification

A trivial detector scores each window by the maximum absolute standardized per-channel window mean,  $\max_{ch} |u|$  ( $u$  denotes the standardized per-channel window mean, distinct from the latent  $z$  of §4.3), the simplest “is any channel out of its normal range” rule. An anomaly window is labeled **easy** if this trivial score exceeds the 99th percentile of train-normal, and **difficult** otherwise. Difficult windows are those this max-mean rule does not separate (a stronger split is tested in §6). We report all methods on three subsets: **Easy**, **Difficult**, and **All**. In every subset the positive class is that subset’s anomaly windows and the negative class is the full set of normal windows, so the subsets differ only in which anomalies they include. The difficult column is the discriminative one.

## 5.4 Comparable metrics

We report two raw, point-wise metrics (each scored window is one evaluation point, with no point adjustment) and are explicit about their comparability across subsets.

- **AUROC** (the probability that a randomly chosen anomaly window outranks a randomly chosen normal window, 0.5 being chance) is prevalence-independent, hence directly comparable across Easy/Difficult/All, and requires no threshold. It is our primary cross-subset metric, reported as a five-seed mean for the learned detectors.
- **F1** is the best raw point-wise F1 over score thresholds (an oracle, field-standard threshold, disclosed), computed per seed and averaged over the five seeds for the

learned detectors. Because the subsets differ in positive count it is not directly comparable across them and serves as a secondary, within-subset figure.

All model fitting and all calibration (density head, nearest-component reference, residual precisions and gate, and every **head standardization**) use train-normal data only. AUROC uses no test-set tuning (all fitting and calibration use train-normal only); only the F1 threshold uses the test-swept oracle, which we flag.

**Significance testing.** Where a difficult-subset margin is called significant, we test the paired difference in seed-averaged difficult-subset AUROC between LatAD and the strongest competing baseline with a bootstrap that respects the serial dependence of the data. Normal windows are resampled by a moving-block bootstrap with a block length of 3 windows on WADI and 4 on HAI and SWaT, exceeding the window-overlap span, so overlapping windows are resampled as blocks; anomalous windows are resampled by an episode-level bootstrap that draws whole attack episodes with replacement and contributes their difficult windows, preserving within-attack dependence. From 2000 resamples we report the 95% confidence interval (CI) of the paired difference by the percentile method (the 2.5th and 97.5th quantiles) and a one-sided  $p$ -value as the fraction of resamples in which the difference is at most zero. These verdicts are robust to the choice of block length. Selecting it instead from the empirical serial dependence, the first lag at which the normal-window score autocorrelation falls below  $1/e$ , gives 264 windows on HAI (whose community score stays correlated under the record's front-loaded drift), 3 on the stationary WADI record, and 4 on SWaT; at this autocorrelation-selected length, and under a seed-hierarchical resample that also resamples the five training seeds, every head-line comparison keeps the same significant-or-tie verdict. Because the paired difference cancels shared normal-window noise, a longer block widens the interval only slightly rather than moving the estimate.

### 5.5 Baselines and deep detectors

Classical baselines are Isolation Forest [34] and a deep autoencoder, both on the same window features, together with a linear cross-channel baseline (LinRes): a leave-one-channel-out linear regression that predicts each channel's window summary from the others, with discrete actuator channels one-hot encoded, scored by its residual. **Baseline settings are listed in Appendix D (Table D2).** For modern SOTA we run USAD [36] and TranAD [37] through the TranAD evaluation harness **over five seeds on all three datasets** and score them with the *same raw point-wise metrics*, aggregating their per-timestep scores onto our window grid. For HAI, the SOTA harness produces scores at a downsampled rate, so we map them to our full-resolution window grid by an integer-ratio correction; our detector and all local baselines use HAI at native resolution. **We also benchmark the graph-based GDN [38] under the identical protocol: its per-timestep anomaly scores are re-aggregated onto our window grid and difficulty subsets exactly as the deep detectors are, giving a single-seed, construct-matched GDN figure on all three datasets (Table 2), scored by AUROC only.** We do *not* apply point adjustment to any method (§2.2). **The trivial max|ul rule of §5.3 is also shown as a diagnostic row.**

## 6 Results

Table 2 reports AUROC and best raw point-wise F1 for every method on the three datasets and the three subsets. The reported model is **LatAD (regime-community)** (§4.4).

**Table 2.** Per-dataset, per-subset results [AUROC, F1] on three real CPS attack benchmarks (higher is better; AUROC uses no test-set tuning, F1 is the best raw point-wise F1 over thresholds). **Isolation Forest, the AutoEncoder, USAD, TranAD, and LatAD report AUROC as mean±standard deviation over five seeds on all three datasets (the standard deviation is omitted where it rounds to 0.000).**

GDN is included as a construct-matched, single-seed SOTA baseline: its per-timestep scores are re-aggregated onto the same window grid and difficulty subsets as every other method (§5.5), and are scored by AUROC only (F1 not computed; shown as “–”). Bold marks the best AUROC in each dataset×subset column (ties bolded jointly). **LatAD (regime-community)** is the reported detector (§4.4); its single-latent **LatAD (global density)** base ablation is reported with the factorization axis in Table A1 and the cost in Table B1. Baselines are defined in §5.5.

| Method                                    | Group    | All         |       | Easy        |       | Difficult   |       |
|-------------------------------------------|----------|-------------|-------|-------------|-------|-------------|-------|
|                                           |          | AUROC       | F1    | AUROC       | F1    | AUROC       | F1    |
| WADI (anom 56 = 26 easy + 30 difficult)   |          |             |       |             |       |             |       |
| trivial max u                             | Baseline | 0.786       | 0.62  | 0.999       | 0.98  | 0.601       | 0.13  |
| Isolation Forest                          | Baseline | 0.725±0.005 | 0.39  | 0.829±0.007 | 0.42  | 0.634±0.006 | 0.17  |
| AutoEncoder                               | Baseline | 0.792±0.002 | 0.55  | 0.981±0.001 | 0.82  | 0.628±0.004 | 0.18  |
| LinRes (one-hot)                          | Baseline | 0.834       | 0.68  | 0.932       | 0.73  | 0.750       | 0.53  |
| USAD                                      | SOTA     | 0.757       | 0.49  | 0.962       | 0.73  | 0.579       | 0.16  |
| TranAD                                    | SOTA     | 0.786       | 0.54  | 0.986       | 0.74  | 0.613       | 0.20  |
| GDN                                       | SOTA     | 0.801       | –     | 0.963       | –     | 0.660       | –     |
| LatAD (regime-community)                  | Ours     | 0.827±0.009 | 0.57  | 0.892±0.009 | 0.58  | 0.771±0.023 | 0.42  |
| HAI (anom 652 = 485 easy + 167 difficult) |          |             |       |             |       |             |       |
| trivial max u                             | Baseline | 0.806       | 0.666 | 0.966       | 0.800 | 0.340       | 0.000 |
| Isolation Forest                          | Baseline | 0.844±0.011 | 0.418 | 0.919±0.016 | 0.468 | 0.627±0.005 | 0.059 |
| AutoEncoder                               | Baseline | 0.923±0.001 | 0.733 | 0.980       | 0.803 | 0.757±0.003 | 0.340 |
| LinRes (one-hot)                          | Baseline | 0.779       | 0.499 | 0.846       | 0.605 | 0.586       | 0.121 |
| USAD                                      | SOTA     | 0.849±0.005 | 0.691 | 0.969±0.001 | 0.823 | 0.497±0.018 | 0.018 |
| TranAD                                    | SOTA     | 0.834±0.002 | 0.692 | 0.968       | 0.824 | 0.445±0.010 | 0.015 |
| GDN                                       | SOTA     | 0.845       | –     | 0.970       | –     | 0.481       | –     |
| LatAD (regime-community)                  | Ours     | 0.948±0.004 | 0.760 | 0.984±0.002 | 0.820 | 0.845±0.018 | 0.360 |
| SWaT (anom 233 = 142 easy + 91 difficult) |          |             |       |             |       |             |       |
| trivial max u                             | Baseline | 0.853       | 0.680 | 0.997       | 0.916 | 0.627       | 0.214 |
| Isolation Forest                          | Baseline | 0.800±0.004 | 0.642 | 0.961±0.004 | 0.855 | 0.550±0.007 | 0.170 |
| AutoEncoder                               | Baseline | 0.786±0.001 | 0.680 | 0.959±0.001 | 0.906 | 0.517±0.003 | 0.141 |
| LinRes (one-hot)                          | Baseline | 0.776       | 0.716 | 0.976       | 0.941 | 0.464       | 0.099 |
| USAD                                      | SOTA     | 0.763±0.001 | 0.669 | 0.946       | 0.907 | 0.477±0.001 | 0.129 |
| TranAD                                    | SOTA     | 0.761±0.001 | 0.669 | 0.943       | 0.908 | 0.477±0.002 | 0.129 |
| GDN                                       | SOTA     | 0.761       | –     | 0.945       | –     | 0.473       | –     |
| LatAD (regime-community)                  | Ours     | 0.788±0.008 | 0.698 | 0.957±0.005 | 0.893 | 0.524±0.024 | 0.238 |

**Overall detection.** LatAD attains the best All-subset AUROC of any learned detector on HAI (0.948), clearing every baseline and all three deep detectors. On WADI it clears every evaluated nonlinear detector on the full set (0.827), with the linear baseline marginally ahead there (0.834 vs 0.827); WADI's advantage for LatAD is on the difficult subset. On SWaT the All-subset scores are compressed and led by the trivial max|u| rule (0.853), a signature of the severe measured drift on that record (§7): the raw community score reaches 0.788, above the deep detectors (0.761–0.763) but below the trivial rule, and the drift-aware typing of §7 lifts it to 0.880. The gain over the single-latent base (LatAD global density) comes from factorizing the density over regime communities (§4.4), and is largest on the difficult subset.

**The difficult subset.** This is the subset the design targets (Figure 3). The clearest demonstration is HAI: LatAD reaches 0.845 on the difficult subset (167 windows spanning 26 attack episodes), well above the AutoEncoder (0.757) and far above the deep detectors USAD and TranAD, which sit near chance. The margin over the AutoEncoder is statistically significant: an episode-block bootstrap of the paired difference (resampling attack

episodes and normal windows, averaged over five seeds) gives +0.088 with a 95% confidence interval of [0.042, 0.157] and  $P(\text{difference} \leq 0) < 0.0005$ .

On SWaT the raw difficult subset (91 windows across 23 attack episodes) is drift-dominated. The raw community score reaches 0.524, above every deep detector (0.473–0.477) and the single latent (0.472) but not separated from the strongest linear baseline (+0.060 over LinRes 0.464, 95% CI including zero). A drift-aware changepoint decomposition of the same community surprises recovers the SWaT difficult subset to 0.723 (§7), a significant +0.200 over the raw score (§7).

On WADI LatAD reaches 0.771 (30 difficult windows over 8 attack episodes), significantly ahead of every nonlinear detector. A per-detector episode-block bootstrap of the paired difference clears each one with  $P < 0.05$ : vs the AutoEncoder +0.143 ( $P = 0.001$ ), USAD +0.192 ( $P < 0.0005$ ), TranAD +0.157 ( $P = 0.0005$ ), Isolation Forest +0.136 ( $P < 0.0005$ ), and GDN +0.111 ( $P < 0.0005$ ). LatAD is not significantly different from the linear baseline LinRes (0.750): +0.020, 95% CI [−0.161, 0.205],  $P = 0.46$  over 8 episodes. The WADI lead is carried by the community density aggregation (cohesion-weighted Higher Criticism over the community experts, 0.795): the single-latent global density scores 0.634 on this difficult subset, so the community factorization rather than the single latent supplies the WADI signal. The tie with the linear baseline is expected from the mechanism: the cross-channel latent density, LatAD's distinctive lever, adds +0.200 on HAI but only +0.017 on WADI (Table A4), so the WADI difficult faults carry no cross-channel-only structure and a linear per-channel model extracts the same signal the density does.

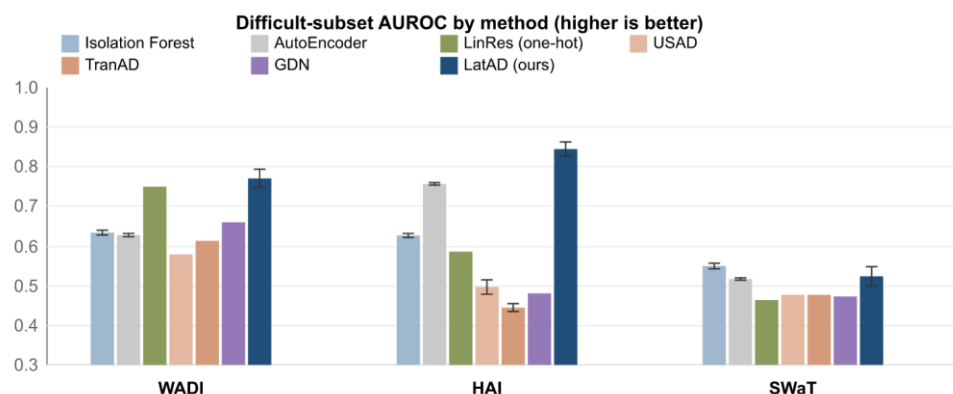
**Validity of the difficulty split.** The trivial  $\max |u|$  rule is strong on Easy by construction (WADI 0.999, HAI 0.966, SWaT 0.997) and drops on the difficult subset (WADI 0.601, HAI 0.340, SWaT 0.627): the difficult subset is precisely the anomalies this univariate  $\max$ -mean rule separates least well, so a method that scores well there is detecting anomalies the univariate rule misses: joint-structure faults on HAI, while some of the WADI residual remains single-channel or linear (which is why the linear baseline stays competitive there). The drop is steepest on HAI, where the rule falls to near chance; on WADI it falls well below the Easy ceiling. On SWaT the rule stays elevated on the difficult subset (0.627), which is itself a drift signature: a non-stationary record lifts late-segment normal windows so that even a univariate rule retains rank, and separating drift from genuine anomaly there is the subject of §7.

**Robustness to a stronger difficulty definition.** The §5.3 split flags a window easy only when the maximum absolute standardized per-channel *mean* exceeds its 99th train-normal percentile, so a window could in principle be “difficult” yet be separable by a different univariate statistic. We therefore repeat the analysis with a stronger split that labels a window easy if the maximum over *all six* standardized per-channel statistics (mean, standard deviation, minimum, maximum, first-to-last difference, and range) exceeds its 99th train-normal percentile, removing any window a single univariate statistic separates. On the resulting HAI difficult subset (110 windows) the single-latent LatAD (global density) still leads every baseline (0.675, versus 0.622 for the linear cross-channel baseline, 0.604 for the AutoEncoder, and 0.528 for Isolation Forest), so the HAI advantage is not an artifact of the particular statistic that defines difficulty.

**The double-hard subset.** As a secondary robustness stress test we further remove, from the difficult subset, every window that a linear principal-component subspace already separates; the filter construction is detailed in Appendix A, beside Table A3. On the resulting double-hard windows (windows / distinct episodes: WADI 29 / 8, HAI 55 / 13, SWaT 28 / 9) LatAD attains the highest AUROC on WADI and HAI (0.763 and 0.648), a numerical lead over the strongest baseline (LinRes) of +0.021 (WADI, 8 episodes) and +0.101 (HAI, 13 episodes). On SWaT the raw density score is drift-dominated on this

subset (0.178; the record carries severe measured drift, §7), and the drift-aware change-point typing of §7 recovers the SWaT double-hard subset to 0.671.

**Deep SOTA under raw metrics.** Re-computed with raw point-wise metrics, the deep detectors USAD and TranAD fall to 0.497 and 0.445 on the HAI difficult subset, near chance and far below LatAD's 0.845, even while they remain strong on Easy (0.94–0.99). On WADI they trail LatAD without collapsing (0.579 / 0.613 versus 0.771). This is the concrete face of the “illusion of progress”: models that look strong under point-adjusted F1 have little difficult-fault signal once the metric is raw, and lose most sharply where the fault breaks joint structure (above), consistent across datasets.



**Figure 3.** Difficult-subset AUROC by method, values as Table 2; whiskers show  $\pm 1$  standard deviation (SD) over five seeds where available. LatAD leads the difficult subset on HAI (0.845, significantly over the AutoEncoder) and leads every nonlinear detector on WADI (0.771), tying the linear baseline; on SWaT the raw difficult scores are drift-compressed near chance (§7). Significance tests are given in the text.

**Source of the gain.** The difficult-subset advantage is driven primarily by cross-channel density modeling rather than by the window representation, which is a separate and smaller complementary lever (Appendix A, Table A2). Replacing the cross-channel latent density with a channel-independent product of per-channel densities (Table A4) isolates where cross-channel structure, rather than per-channel marginals, carries the signal: the cross-channel density adds +0.200 on HAI over the channel-independent baseline, while on the WADI difficult subset it adds essentially nothing (+0.017). WADI's difficult anomalies are single-channel or linear, so per-channel marginals already suffice (excluded channel: §5.1). On SWaT the raw cross-channel density is drift-dominated (§7) and does not exceed the marginal product (−0.050). The head-level decomposition (Table A5) and the factorization/aggregation ablation (Table A1) localize the same gain: the latent density anchors the base score, and factorizing it over correlation communities lifts the difficult subset further, most on WADI and on HAI (Appendix A).

## 7 Drift spectrum and drift-aware anomaly typing

Everything that departs from training-normal is an anomaly, but departures come in two physically distinct kinds: an abrupt *change-point*, where the system jumps to an off-normal state, and a gradual *drift*, where the normal operating point itself migrates over the record as components age or recalibrate. A snapshot detector scores both as surprise, so on a non-stationary record the accumulating drift of normal operation swamps the sharp signature of a genuine anomaly. This is the mechanism behind the compressed SWaT scores of §6, and it is a known obstacle for time-series detectors trained on a stationary normal [54, 55]. This section measures the drift each benchmark carries (§7.1) and shows that separating

the changepoint component from the drift component recovers detection under drift while leaving stationary records unchanged (§7.2).

This admits two readings of drift, and we are explicit about which we adopt. *Formally*, drift is out-of-distribution: the operating point has migrated off the training support, so a strict “an anomaly is any departure from training-normal” detector must flag all of it, which is why the untyped SWaT normal-window false-alarm rate reaches 79%. *Operationally*, we adopt an inductive bias: slow, smooth, plant-wide migration of the operating point, the signature of benign component aging, recalibration, or seasonal shift, is not the fault an operator wants paged, so we treat it as a permissible extension of normal and score only fast departures from the drifting baseline. The drift-aware typing of §7.2 encodes this prior causally, decomposing each community surprise into a tolerated slow baseline and a scored fast residual rather than reclassifying drift as benign by fiat; setting the drift timescale to the whole record removes the bias and recovers the strict formal detector exactly.

### 7.1 The drift spectrum

We measure drift directly as the fraction of test-normal windows whose per-window mean community surprise exceeds the train-normal 99th percentile. The three benchmarks span a wide range: WADI 1.5%, HAI 39%, and SWaT 69%. The ordering is physically grounded. WADI's two-day test record leaves no time for component aging, so its normal is effectively stationary. HAI carries a moderate, front-loaded drift: a decaying level offset in two valve-position and turbine communities that begins early in the test record and relaxes over it. SWaT is a single continuous eleven-day run of a live water-treatment plant, and its drift is large, monotone, and localized to the process analyzers and the reverse-osmosis train: the drifting communities are exactly those built from the analyzer channels (AIT2xx/4xx/5xx) and the RO flow and pressure channels (FIT4xx/5xx, PIT5xx; the AIT, FIT, and PIT families are analyzer, flow, and pressure-transmitter tags), matching the physical signature of reverse-osmosis membrane fouling and analyzer electrode drift, so the per-community drift map is itself a physically-grounded degradation-localization signal for condition monitoring. Datasets recorded as a single long run over non-stationary physical processes therefore carry drift that discrete-attack testbeds with short test records do not, a distinction consistent with the recalibration handling documented for water-treatment analyzer channels elsewhere [53] and with the clipping of recalibrated analyzer channels described in §5.1.

### 7.2 Drift-aware changepoint typing

To separate the two kinds of departure we decompose each community's surprise causally, using only past windows. A *slow* on-trend baseline is the rolling median of the surprise over the previous 24 h of wall-clock, and a *fast* residual is the surprise minus that baseline, divided by the larger of the rolling median absolute deviation over the same window and the train-normal (calibration) deviation, so no community is ever more sensitive than at training time. A window is typed *changepoint* when its fast residual exceeds its calibration 99th percentile, and *drift* when the slow baseline alone is off the training distribution. Scoring the changepoint component with the same cohesion-weighted Higher Criticism combiner (§4.4) gives the drift-aware detector; the 24 h timescale is fixed a priori from the plant's slowest control loop, and setting the window to the whole record reproduces the headline score exactly.

Table 3 reports the effect. On SWaT the changepoint score lifts the community fusion from the drift-dominated 0.524 difficult and 0.178 double-hard to 0.723 and 0.671 (All 0.788 to 0.880). Against the reported score the gain is significant on every subset: difficult +0.200 (95% CI [0.018, 0.355],  $P = 0.014$ ), double-hard +0.493 (95% CI [0.349, 0.619],  $P < 0.001$ ), All +0.092 (95% CI [0.019, 0.301],  $P = 0.006$ ). Against the strongest clean

alternative on SWaT, the trivial  $\max|u|$  rule, it leads the difficult subset by +0.096 ( $P = 0.031$ ) and the double-hard subset by +0.157 ( $P = 0.036$ ); the deep detectors and GDN sit far below both. The normal-window false-alarm rate at the train-calibrated threshold falls from 79% to 6.5%. At the same threshold the attack detection rate is 0.687 overall and 0.330 on the difficult subset (against 0.918 and 0.791 at the untyped threshold that fires on 79% of normal windows), so the typing trades a small loss of recall for a large reduction in false alarms; the recovery is not sensitive to the 24 h baseline (difficult-subset AUROC 0.757, 0.723, and 0.706 at 12, 24, and 48 h). On WADI (stationary) and HAI (moderate, front-loaded drift), the same decomposition at the same 24 h timescale is a near-no-op: HAI difficult 0.845 to 0.835 (within seed variation; Table 2) and WADI difficult 0.771 to 0.779, both within the bootstrap interval.

**Table 3.** Drift-aware changepoint typing: AUROC of the reported community score and of its changepoint-typed component (the fast residual scored by the same combiner), five-seed mean, on the All, Difficult, and double-hard subsets. The typing is a near-no-op on WADI (stationary) and HAI (moderate, front-loaded drift) and recovers the drift-dominated SWaT record; SWaT significance against the reported score and against the trivial rule is given in the text.

| Dataset | All (reported $\rightarrow$ typed) | Difficult (reported $\rightarrow$ typed) | Double-hard (reported $\rightarrow$ typed) |
|---------|------------------------------------|------------------------------------------|--------------------------------------------|
| WADI    | 0.827 $\rightarrow$ 0.830          | 0.771 $\rightarrow$ 0.779                | 0.763 $\rightarrow$ 0.771                  |
| HAI     | 0.948 $\rightarrow$ 0.943          | 0.845 $\rightarrow$ 0.835                | 0.648 $\rightarrow$ 0.636                  |
| SWaT    | 0.788 $\rightarrow$ <b>0.880</b>   | 0.524 $\rightarrow$ <b>0.723</b>         | 0.178 $\rightarrow$ <b>0.671</b>           |

The typing validates against the only available ground truth, the attack labels, used here for evaluation only. Of the windows the reported detector flags as anomalous, 74.8% on SWaT, 88.5% on HAI, and 61.9% on WADI type as changepoint; of the flagged normal windows that carry drift, 97.4% on SWaT and 92.5% on HAI type as drift, and on WADI the drifted-normal fraction is negligible (1.5%, Table 4). Attacks type as changepoint and drifting normals as drift, which is the pattern the decomposition is built to produce.

**Table 4.** Drift spectrum and typing validation. *Test-normal drifted* is the fraction of test-normal windows whose per-window mean community surprise exceeds the train-normal 99th percentile. *Attacks typed changepoint* and *drifted normals typed drift* use the attack labels for evaluation only. The false-alarm column is the normal-window flag rate at the train-calibrated threshold, reported then changepoint-typed.

| Dataset | Test-normal drifted | Normal false alarm (reported $\rightarrow$ typed) | Flagged attacks typed changepoint | Flagged drifted normals typed drift |
|---------|---------------------|---------------------------------------------------|-----------------------------------|-------------------------------------|
| WADI    | 1.5%                | 1.5% $\rightarrow$ 1.2%                           | 61.9%                             | n/a (1.5% drifted)                  |
| HAI     | 39%                 | 10.5% $\rightarrow$ 7.2%                          | 88.5%                             | 92.5%                               |
| SWaT    | 69%                 | 79% $\rightarrow$ 6.5%                            | 74.8%                             | 97.4%                               |

Three caveats apply. First, ranking recovers but a deployable absolute alarm threshold in a heavily drifted recording segment (a contiguous stretch of the test record) does not follow automatically: the rolling scale deflates the whole late SWaT record, so a fixed train-calibrated threshold under-fires there even though the late attacks rank at the 94th to 98th percentile against their contemporaneous normals; a heavily drifted deployment adds a rolling calibration on top of the typing, which this study does not report. Second, ranking each attack only against normal windows from the same stretch of the record, two of SWaT's 25 windowed attack episodes (24 with a published target, 23 contributing difficult windows) drop by more than 0.10 (one of them within the 24 h shadow of a 10 h attack whose residuals inflate the rolling deviation of the mechanical communities), while five rise; HAI and WADI show no drop. Third, an injected synthetic ramp is largely

removed by the causal baseline, but a one-sided rolling median lags a ramp by half its window, which leaves a residual bias on WADI's two-day test record (twice the baseline window); the effect is a no-op in practice because WADI carries no real drift.

## 8 Discussion

**Summary of findings.** Under a raw-metric, difficulty-stratified protocol with train-normal-only calibration, LatAD leads the difficult subset on HAI (0.845, significantly) and leads every nonlinear detector on WADI (0.771), tying the linear baseline, while the deep detectors USAD and TranAD fall near chance on HAI (0.44–0.50); on WADI the community factorization, not the single latent, carries the lead (§6). On SWaT the raw scores are drift-dominated, and the drift-aware changepoint typing of §7 recovers the difficult subset to 0.723 and the double-hard subset to 0.671. Modeling normal as a jointly learned latent plus explicit regime clustering, scored by probability and factorized over the plant's subsystems, is what delivers the advantage on WADI (stationary) and HAI (moderate, front-loaded drift); a causal drift/changepoint decomposition extends it to the severely non-stationary SWaT record.

**Why reconstruction fails and density succeeds.** The deep detectors USAD and TranAD score by reconstruction. On multimodal normal data reconstruction and probability diverge: a flexible decoder trained to reproduce several regimes interpolates between them, so an improbable between-regime or in-regime-tail combination sits inside the decoder's reachable range and reconstructs with low error. Scoring by density rather than reconstruction is the right default: the latent-density head alone, with no residual term and no factorization, beats USAD and TranAD on the HAI and WADI difficult subsets (Table A5 vs Table 2), where the faults are reconstructable but improbable; on SWaT the raw difficult subset is drift-dominated (§7): every snapshot score, whatever its head, is compressed to the drift floor, and the drift-aware typing of §7 resolves it (0.723). On HAI the reconstruction detectors place the difficult anomalies (labeled attacks) at the median of normal operation, whereas the density head places them at the 93rd percentile; 27 percent of HAI's difficult anomalies reconstruct at least as well as a typical normal window yet fall above the 90th normal percentile in density. On WADI that share is zero, and 70 percent of the difficult anomalies are weak under both scores, which is why the community factorization rather than the head carries the WADI gain. Figure A1 illustrates this reachability-versus-probability crossover on synthetic data. A reconstruction residual *whitened on train-normal* is nonetheless a stronger term than the reconstruction the deep detectors score with, which is why LatAD keeps it as a gated head (Table A5) where it generalizes to held-out normal instead of discarding reconstruction. The base score therefore omits reconstruction and scores in the jointly learned latent, with a high-resolution density head for non-Gaussian fringes (A3) and a rare-regime-safe nearest-component likelihood (A5).

**Why local density outperforms global density.** A local fault is diluted by a single global density but concentrated by a per-community one (§4.4). This is the mechanism behind the WADI gain: a coordinated deviation (shift) confined to a small correlated subsystem is improbable within that subsystem's own density yet unremarkable in the whole-plant density, so factorization lifts the WADI difficult-subset score (Table A1: 0.634 → 0.795).

Table A5 decomposes the score into its heads across five seeds and all three datasets. On HAI reconstruction is informative but the weakest base head (0.695) while the latent density dominates (0.802); on WADI the standalone reconstruction residual (0.740) fails the held-out-normal generalization gate (§4.3 iii) and is switched off, so the reported single-latent model relies on the latent density (0.656) and the density-plus-nearest base (0.634). Where the gate turns the residual on, on HAI and SWaT, its held-out-normal

generalization passes; on HAI it lifts the base score (0.801→0.820). On SWaT every raw head sits at the drift-suppressed difficult-subset floor (0.47–0.57) because the record carries severe drift (§7): reconstruction is the strongest single head (0.567) and the residual gate is on, yet the base and base-plus-residual scores are both 0.472, so the SWaT difficult subset is recovered by the changepoint typing of §7, not by any snapshot head. The decomposition supports the mechanism at the head level: the latent density anchors the base score, and reconstruction is added back only where it generalizes to held-out normal.

**Assumption A8 (between-regime overlap).** A8 posits regimes lying so close in observation space that a window can be improbable while every channel stays in range. It has two failure modes: over-coverage (a density head so coarse that its normal mass spreads across the gap between regimes and covers an isolated anomaly) and masking (an anomaly of one regime absorbed by an adjacent regime's density). Its precondition, adjacent-but-distinct regimes, holds on every multi-loop plant we screened (WADI, HAI, SWaT, Cranfield, MetroPT [56], and the SMD server machines [31]: 94 to 100 percent of train-normal mass sits in a discovered regime with a valley-separated neighbor within twice its own normal envelope). The heavy overlap itself is absent: under four observation-space measures (the overlap coefficient is tabulated in Table C1, alongside its close-distinct-mass, masking-rate, and isolated-anomaly over-coverage columns) no benchmark approaches the 30-percent-overlap positive control, and the labeled anomaly-masking rate never exceeds its lag-matched regime-switch base rate (Appendix C, Table C1). The over-coverage mode is already defended by the reported model: a wide two-component head misses about 80 percent of WADI's isolated anomalies, the 80-component density head of §4.3 (i) about 13 percent. The masking mode is identifiable only through the path by which a window arrived, so it belongs with A10 and is left to operational-scale data (Appendix C). The reported model carries no dedicated overlap head.

**Density generalizes the linear residual.** The WADI tie has a precise reading. Under a single Gaussian the negative log-density is the whitened (Mahalanobis) residual energy of Equation (6), and a leave-one-channel-out linear regressor (LinRes) reads off exactly that inverse covariance; LinRes is therefore the single-full-covariance-Gaussian special case of a density detector, not a different kind of method. LatAD generalizes it in two directions, a mixture over regimes and a nonlinear latent encoding, so it reduces to the linear residual when normal is unimodal with linear channel relations and separates from it only to the degree normal departs from a single linear Gaussian. The cross-channel latent density accordingly adds +0.200 on the multimodal HAI record but +0.017 on the near-single-mode WADI record (silhouette 0.06; Table A4): LatAD wins on HAI and correctly degenerates to the linear baseline on WADI. The auto-gated residual head (§4.3 iii) is this Gaussian-density term made explicit, and the mixture-density and community factorization are its multimodal generalization.

**Benchmark coverage as a diagnosable property.** The per-community model also flags where a benchmark leaves training coverage thin: on HAI one operating regime is almost absent from the training record (0.4% of train windows against 6.3% of test) and appears as a contiguous block of test-normal windows after training ends, which LatAD's own regime model flags as an unseen regime and where its 5%-budget false alarms concentrate, a recording-duration limit rather than a fault.

**Trajectory assumptions A9–A10 and benchmark scale.** The reported model is window-only and leverages the instantaneous-structure assumptions A1–A7 (A8 is specified but not realized). A9 (multiscale, many clocks) is present in the data as a measured property: per-channel autocorrelation timescales span 3.4 to 3.7 decades on all three plants. It is only partly exploitable at the window scale here. On HAI a longer 600-sample window adds about +0.06 difficult-subset AUROC on the part of the difficult subset whose history is attack-free, a slow persistent fault that accumulates over the longer horizon,

corroborating the +0.071 that explicit temporal features add in Appendix A; on WADI and SWaT longer windows do not help. A10 (path dependence) shows no evidence under these tests rather than merely unobserved: three conditional scorers (a linear autoregressor on the two preceding windows, a nearest-neighbor successor model, and a regime-transition matrix), validated on a synthetic positive control and guarded by an order-shuffle control, find no window that is marginally normal yet improbable given its history above the shuffle floor, on all three benchmarks, once windows whose history already contains an attack are excluded. “Snapshot-detectable” is thus a measured finding for these benchmarks, not an assumption. A8 through A10, and the rare-regime safeguard of A5, register as absent, inert, or unexercised on the public CPS benchmarks precisely because those benchmarks are small and single-mode: days of data, one nominal operating mode, and discrete abrupt attacks. These are properties of the available data, not of the method; the operational-scale records that would exercise each of them (many imbalanced regimes for A5 and A8, long duration for A9, and history-dependent run-to-failure trajectories for A10) are set out in §2.5, and curating a benchmark that genuinely exhibits heavy-tailed, discrete multi-regime structure remains the main barrier to exercising the full MIIM assumption set.

**Alternative realizations and validation.** The MIIM assumptions fix the structure to model rather than the estimator, so the detector of §4 is one point in a larger design space. The regime-community realization (§4.4) exercises the factorization and sparsity-adaptive-combination routes; further routes remain open within the same assumption set: non-parametric or normalizing-flow densities per community, alternative unsupervised partitioners, and a deeper multiscale community hierarchy than the current nested-subtree construction. Because A7 spans discrete and mixed-type channels, native categorical likelihoods for discrete channels (HI-VAE [57], VAEM [58]) and graphical-model ICS detectors that type actuator states (TABOR [59]) are complementary realizations of A7 that this instantaneous, all-Gaussian model does not use. All community-construction and combiner settings are shared across datasets and not tuned per dataset (§5.2). Validating the regime-community realization on additional CPS systems is the next test; because its build and calibration use train-normal data only, that test requires no labels.

**Operator-facing output.** The three practitioner-facing properties compose into one deployment story, using only train-normal statistics and online per-window scoring. A site is commissioned from its own normal telemetry with no labeled attacks (Figure 1); each alarm is localized to a physical subsystem via a three-community shortlist ranked by train-normal p-value (Table 5); and on a drifting record the drift-aware typing of §7 holds the normal-window false-alarm rate to 6.5 percent, down from 79 percent untyped, while recovering detection. All three run within the edge training and inference budgets of Appendix B.

**Subsystem localization.** A three-community shortlist ranked by train-normal p-value contains the attacked subsystem in 68 percent of HAI attacks, against 56 percent for a random shortlist of the same size (Table 5). The ranking is the one the Higher Criticism combiner already computes: communities are ordered by their train-normal upper-tail p-value (communities whose calibration surprise is degenerate carry no p-value and are not ranked), and the top community is named once per attack episode. On HAI that single named community contains an attacked channel in 39 percent of attacks (15 of 38), against a coverage-aware random baseline (the chance that a randomly named community of the same size contains an attacked channel) of 25 percent ( $P = 0.024$ ). The named community averages six channels of 59. WADI shows the same pattern (top-1 0.33 versus 0.13 random, top-3 0.56 versus 0.31), directionally consistent with HAI ( $P = 0.086$  over 9 attacks). Under its measured drift (§7) the train-normal community ranking is uninformative on SWaT, so localization is reported for HAI and WADI. The attack targets are read from the published

per-testbed attack tables [50, 51, 52]. Two limits apply: a community whose surprise drifts above its calibration range is named disproportionately often, so the pointer is a triage shortlist rather than a diagnosis; and because larger communities are easier to hit, the named-community size is reported and the top-3 headline is stated against the coverage-aware baseline.

**Table 5.** Subsystem localization at the attack-episode level, rank rule described in the text above. *Top-1* names the single most-surprised community; *top-3* is a three-community shortlist. The random baselines are coverage-aware (the chance that a randomly named community of the same resolution contains an attacked channel), averaged over episodes. Bold marks HAI, where the top-1 localization margin is significant ( $P = 0.024$ ). SWaT is omitted: its train-normal ranking is defeated by the record's drift (§7).

| Dataset | Attack episodes<br>(targets known) | Named-community<br>size (channels /<br>plant) | Top-<br>1   | Random<br>top-1 | Top-<br>3   | Random<br>top-3 |
|---------|------------------------------------|-----------------------------------------------|-------------|-----------------|-------------|-----------------|
| HAI     | 38 (36 coverable)                  | 5.6 / 59                                      | <b>0.39</b> | 0.25            | <b>0.68</b> | 0.56            |
| WADI    | 9 (7 coverable)                    | 5.4 / 122                                     | 0.33        | 0.13            | 0.56        | 0.31            |

Pushing inference to the network edge (a gateway or the device itself) reduces telemetry backhaul and enables local response, but the regime-community realization instantiates one density model per community, so its inference cost grows with the number of subsystems. Where that footprint matters, adaptive-inference optimizations are a natural fit: early-exit cascades terminate computation early on inputs that are already confidently normal and cut average inference cost substantially, for example CalxNet [60], reported to reduce inference cost by roughly 30 to 70 percent. Training and inference cost are in Appendix B (Table B1); CalxNet is a further-reduction route on top of those figures for the per-community stack on constrained edge hardware.

**Limitations.** Three boundary conditions apply: three datasets, one trained configuration per dataset (evaluated over five seeds), and a single instantaneous representation. Two further boundaries follow from the data: WADI results are on the 122-channel set with the rescaled analyzer channel removed (§5.1), and HAI's test record contains one operating regime almost absent from training; all reported numbers use the full test set nonetheless. AUROC is threshold-free and reported as a five-seed mean; the F1 column is a best-threshold raw F1. The easy/difficult split is a property of one cut, the trivial detector's 99th-percentile-of-train-normal threshold; a different percentile would redraw the boundary between the subsets, though the gap between the trivial rule's Easy and Difficult performance is large enough that the split is not knife-edge. We report only raw point-wise metrics; no point adjustment is applied to any method, so the reported numbers are lower than point-adjusted leaderboard figures.

## 9 Conclusion

We restated the structural assumptions of CPS normal data as an explicit MIIM list (A1–A10) and built LatAD, a detector that models normal as a jointly learned latent plus explicit regime clustering, scores anomalies by probability in that latent rather than by reconstruction, and factorizes the density over the plant's correlation-community subsystems, and added a causal drift/changepoint typing that separates a genuine anomaly from the migration of normal itself. Under a raw-metric, difficulty-stratified protocol on three real CPS attack benchmarks, LatAD leads the difficult subset on HAI, with a statistically significant margin over the strongest baseline (paired 95% CI [0.042, 0.157]) while the deep detectors fall near chance, and on WADI (statistically tied with the linear baseline). On the drift-dominated SWaT record the drift-aware changepoint typing recovers the difficult subset to 0.723 (+0.200,  $P = 0.014$ ).

Treating multimodal normal as a single blob, or scoring it by reconstruction alone, misses the reconstructable-but-improbable faults, starkest on HAI, where the deep detectors fall to chance; where a fault does leave a reconstruction signal, the regime-conditioned residual is competitive and LatAD gates it back in.

These results position normal-only latent-density modeling as an anomaly-monitoring approach for multivariate IIoT telemetry, particularly where labels are scarce and anomalous behavior is local to a coupled physical subsystem. Two properties make the detector well-suited to condition monitoring and predictive maintenance: scoring against the nearest regime keeps rare-but-valid states (startups, rare maneuvers) from paging an operator, and the regime-community density catches in-envelope, subsystem-local faults that stay inside every alarm limit. Extending LatAD to the trajectory assumptions (A9, A10, and the masking form of A8 coupled to A10) and validating the regime-community realization on additional CPS systems are the natural next steps.

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**Data Availability Statement:** This study uses only publicly available community benchmark datasets and created no new data. WADI and SWaT are available from iTrust, Singapore University of Technology and Design (<https://itrust.sutd.edu.sg/>, access granted on request); HAI is available from its public release by ETRI. The trained model checkpoints (one per dataset) and the source code supporting the reported results are openly archived on Zenodo at <https://doi.org/10.5281/zenodo.21821524>. Each checkpoint contains learned parameters only, not the underlying data; each is trained with a fixed seed and reproduces that seed's scores, and the released scripts reproduce the five-seed aggregates reported in the tables.

**Conflicts of Interest:** The authors declare no conflicts of interest.

## Appendix A. Ablation over the design axes

LatAD is one point on each design axis of §4: representation, density estimation (single global versus a regime-community ensemble, itself construction plus aggregation), scoring heads, and calibration. Table A1 varies the density-estimation axes on the difficult subset, holding the rest fixed. Factorizing the global density over regime communities lifts WADI and HAI, most on WADI, whose difficult faults are confined to small correlated subsystems that a global density dilutes. On WADI the community aggregation carries the whole effect: cohesion-weighted Higher Criticism over the per-community experts lifts the single-latent global density from 0.634 to 0.795, and fusing in the whole-plant expert (the headline fusion) costs a little, so the fused headline settles at 0.771. On HAI the whole-plant-expert fusion instead lifts the score (0.801 to 0.845), where the hard valve and flow deviations are correlation-conditional. On SWaT the raw aggregations sit near the drift-suppressed floor (0.47 to 0.60); the SWaT difficult subset is resolved by the change-point typing of §7, not by the combiner choice. Among the sparsity-adaptive combiners, cohesion-weighted Higher Criticism is the one that leads on WADI and HAI (all aggregation rules here fused with the whole-plant expert), because it weights a violation by the coupling strength of the subsystem it occurs in (A7) while adapting to the fault's extent.

**Table A1.** Design-axis ablation, difficult-subset AUROC (five-seed mean). Top block: the factorization axis (single global density, then the regime-community ensemble **without and with the whole-plant-expert fusion**). Bottom block: the aggregation axis over the full-stack experts (how per-community surprises are combined). Every aggregation-block row is fused with the **whole-plant** expert: the first four sum the community combiner and the **whole-plant** expert in train-normal **standardized** units (the headline fusion), while the last fuses unweighted Higher Criticism with the **whole-plant** expert by a maximum instead. The reported detector is the full per-community, cohesion-weighted Higher Criticism row. **Bold marks the column maximum on WADI and HAI.**

| Ablation                                                                              | WADI  | HAI   | SWaT  |
|---------------------------------------------------------------------------------------|-------|-------|-------|
| <b>Factorization (density-estimation axis)</b>                                        |       |       |       |
| LatAD (global density), no factorization                                              | 0.634 | 0.811 | 0.472 |
| LatAD (regime-community), cohesion-weighted HC, no whole-plant-expert fusion          | 0.795 | 0.801 | 0.523 |
| <b>LatAD (regime-community), cohesion-weighted HC + whole-plant expert (headline)</b> | 0.771 | 0.845 | 0.524 |
| <b>Aggregation (over full-stack experts)</b>                                          |       |       |       |
| sum (of community surprises)                                                          | 0.698 | 0.827 | 0.595 |
| max (most-surprised community)                                                        | 0.712 | 0.809 | 0.492 |
| Higher Criticism (unweighted)                                                         | 0.760 | 0.793 | 0.513 |
| <b>cohesion-weighted Higher Criticism (headline)</b>                                  | 0.771 | 0.845 | 0.524 |
| Higher Criticism with whole-plant expert, max-fused                                   | 0.749 | 0.770 | 0.468 |

A second check on the representation axis asks whether the advantage comes from the six summary statistics per window rather than from the latent-density mechanism. Table A2 re-runs the single-latent global-density detector with a richer ten-feature temporal/spectral representation (adding within-window slope, velocity mean and standard deviation, a spike measure, and low- and high-band spectral power) in place of the six statistics, at a fixed configuration and a fixed difficulty mask so that only the representation changes; the detector stays a latent-density detector throughout, which isolates the representation from the scoring mechanism. The richer representation is immaterial on WADI and improves the global-density score on HAI (+0.071). On the drift-dominated SWaT record it adds substantially (+0.241), because the temporal and spectral features partly capture the same non-stationarity that §7 models directly; the reported model uses static features throughout for a controlled cross-dataset comparison. On HAI and WADI the representation is a complementary lever aligned with the multiscale-temporal direction A9, not the source of the reported advantage; its integration is left to future work. Feeding the unreduced raw window instead (60 samples per channel, no statistical reduction) strictly lowers the difficult-subset AUROC, by 0.068, 0.230, and 0.060 on WADI, HAI, and SWaT (five seeds), so the advantage is the density mechanism on a compact representation rather than raw information the six statistics withhold.

**Table A2.** Feature-representation check, difficult-subset AUROC of the single-latent global-density detector under the six summary statistics per window versus a ten-feature temporal/spectral set, at a fixed configuration and difficulty mask (five-seed mean±standard deviation). This is a self-contained within-study comparison isolating the representation axis; the absolute values are the single-latent LatAD (global density) base ablation on the difficult subset, not the headline regime-community numbers of the main results table (Table 2).

| Representation                         | WADI        | HAI         | SWaT        |
|----------------------------------------|-------------|-------------|-------------|
| six summary statistics (used by LatAD) | 0.634±0.011 | 0.743±0.011 | 0.247±0.026 |
| ten-feature temporal/spectral          | 0.626±0.017 | 0.814±0.007 | 0.488±0.026 |
| temporal – statistics                  | −0.008      | +0.071      | +0.241      |

Tables A3–A5 collect supporting ablations and the robustness subset referenced from the main text: the double-hard robustness subset (Table A3), the source of the cross-channel gain (Table A4), and the per-head decomposition (Table A5).

The double-hard filter fits a PCA model on train-normal and keeps a difficult window only if it exceeds neither the Hotelling  $T^2$  in-subspace statistic nor the squared prediction (SPE) residual out-of-subspace statistic at their 99th train-normal percentiles: a standard linear-subspace filter that depends only on train-normal, so the detector under test plays no role in constructing the subset, and, unlike a filter built from the leave-one-channel-out linear predictor, it leaves that predictor (LinRes) free to appear as a scored detector in its own right. A boosted leave-one-channel-out linear predictor, fit to squeeze the most from the linear structure, reaches only 0.649 / 0.313 / 0.117 on the three datasets, so boosting the raw linear predictor does not recover it; the drift-aware typing of §7 lifts multiple method families, including this boosted predictor (double-hard 0.117 to 0.553), and LatAD retains the higher post-typing score (0.671).

**Table A3.** Double-hard subset AUROC (five-seed mean±standard deviation for Isolation Forest, the AutoEncoder, USAD, TranAD, and the two LatAD configurations; GDN and the boosted linear predictor are single-seed; the standard deviation is omitted where it rounds to 0.000). The double-hard subset keeps only difficult-subset windows that a linear PCA subspace does not separate (neither the Hotelling  $T^2$  nor the SPE statistic exceeds its 99th train-normal percentile), a standard linear-subspace filter fixed on train-normal only and independent of the detector under test. Windows / distinct attack episodes: WADI 29 / 8, HAI 55 / 13, SWaT 28 / 9. Bold marks the highest AUROC in each column. On SWaT the raw scores are drift-dominated; the drift-aware typing of §7 recovers the LatAD (regime-community) SWaT double-hard AUROC to 0.671.

| Method                          | WADI               | HAI                | SWaT               |
|---------------------------------|--------------------|--------------------|--------------------|
| trivial max u                   | 0.591              | 0.336              | 0.514              |
| Isolation Forest                | 0.622±0.006        | 0.435±0.023        | 0.326±0.012        |
| AutoEncoder                     | 0.616±0.004        | 0.418±0.008        | 0.154±0.006        |
| LinRes (one-hot)                | 0.742              | 0.547              | 0.153              |
| USAD                            | 0.565              | 0.351±0.018        | 0.125±0.004        |
| TranAD                          | 0.601              | 0.296±0.004        | 0.119±0.001        |
| GDN                             | 0.648              | 0.354              | 0.115              |
| boosted LinRes (leave-one-out)  | 0.649              | 0.313              | 0.117              |
| LatAD (global density)          | 0.623±0.012        | 0.559±0.040        | 0.184±0.022        |
| <b>LatAD (regime-community)</b> | <b>0.763±0.024</b> | <b>0.648±0.039</b> | <b>0.178±0.011</b> |

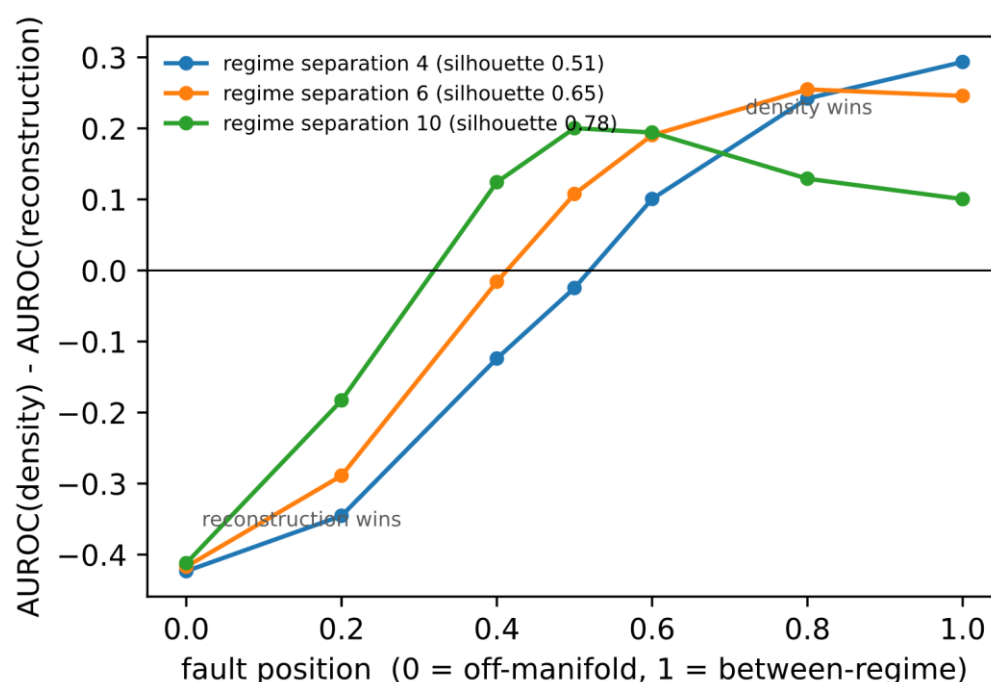
**Table A4.** Source of gain: difficult-subset AUROC (five-seed mean±standard deviation) of the cross-channel latent density against a channel-independent product of per-channel densities, on the WADI, HAI, and SWaT difficult subsets (30 / 167 / 91 anomaly windows). Bold marks HAI, where the cross-channel density adds the largest difficult-subset margin; on WADI it adds essentially nothing (single-channel or linear faults), and on SWaT the raw cross-channel density is drift-dominated. The latent-density values here come from a dedicated source-of-gain run co-computed with the marginal product in one pass; the HAI value (0.789) agrees with the separate head-decomposition run of Table A5 (0.802) within seed variance (both on the same 167-window HAI difficult subset).

| Density model                            | WADI          | HAI           | SWaT          |
|------------------------------------------|---------------|---------------|---------------|
| cross-channel latent density             | 0.656±0.016   | 0.789±0.018   | 0.472±0.020   |
| channel-independent product of marginals | 0.639±0.004   | 0.589±0.004   | 0.522±0.003   |
| <b>cross-channel gain</b>                | <b>+0.017</b> | <b>+0.200</b> | <b>−0.050</b> |

**Table A5.** Difficult-subset AUROC by score head, five-seed mean±standard deviation on all three datasets (a five-seed run). *recon* is the whitened reconstruction residual (the term the base score omits); *latent density* is the  $M = 80$ -component density head of Equation (4) and *nearest-component NLL* is the head of Equation (5); *base* is density+nearest; the final row adds the auto-gated residual head, which is by definition the single-latent *LatAD* (*global density*) configuration and agrees with its Table A1 row within seed variation; its per-community application (§4.4) gives the reported detector. **Bold marks the LatAD (global density) configuration, not a column maximum.**

| Score head                                          | WADI        | HAI         | SWaT        |
|-----------------------------------------------------|-------------|-------------|-------------|
| reconstruction residual (dropped term)              | 0.740±0.002 | 0.695±0.004 | 0.567±0.004 |
| latent density                                      | 0.656±0.016 | 0.802±0.011 | 0.472±0.020 |
| nearest-component NLL                               | 0.603±0.010 | 0.797±0.008 | 0.479±0.025 |
| base (density + nearest)                            | 0.634±0.011 | 0.801±0.010 | 0.472±0.019 |
| <b>base + resid (auto) = LatAD (global density)</b> | 0.634±0.011 | 0.820±0.010 | 0.472±0.019 |

**Figure A1** closes the appendix with a synthetic illustration of the reachability-versus-probability mechanism discussed in §8: as a fault moves from off-manifold to between-regime, density overtakes reconstruction, and the crossover comes earlier for better-separated regimes.



**Figure A1.** Synthetic illustration of the reachability-versus-probability mechanism. On toy data with five Gaussian regimes, the difficult-subset gap between a latent-density score and a reconstruction score (AUROC density minus AUROC reconstruction) is plotted as a fault moves from off-manifold (reconstructable and caught by reconstruction) to between-regime (reconstructable but improbable), at three regime separations. Density overtakes reconstruction as faults become more in-envelope, and the crossover is earlier for better-separated regimes. This is a synthetic mechanism illustration; the benchmark results are in Section 6.

## Appendix B. Computational cost

Table B1 reports training wall-clock and per-window inference latency for LatAD and the deep baselines, benchmarked on one Modal A10G GPU (24 GB) plus a Modal 8-vCPU CPU edge proxy, the same hardware for every method. In training, LatAD (global density)

fits in 3.2, 16, and 3.5 s on WADI, HAI, and SWaT, against 231, 898, and 163 s for TranAD and 96, 592, and 95 s for USAD, so LatAD trains roughly 47 to 72 times faster than TranAD. Per-window GPU latency is 0.7 to 13 ms for the global model and 17 to 32 ms per window (batch 1; up to about 52 ms at batch 256) for the full regime-community stack, against 0.15 ms for a plain autoencoder; a CPU edge proxy scores a window in 0.6 to 14.6 ms (global) and 14.5 to 25 ms (community). The models are small (100 to 209 k parameters global, 0.78 to 1.25 M community) and run within about 3.5 to 3.9 GB of host memory and tens of MB of GPU memory (Table B1, peak 19 to 29 MB for LatAD), placing both configurations inside gateway-class real-time budgets.

**Table B1.** Training and inference cost, per dataset (WADI / HAI / SWaT), measured on one Modal A10G GPU (24 GB) plus a Modal 8-vCPU CPU edge proxy (the same hardware for every method). *Train s* is training wall-clock; *ms/win* is per-window latency at batch 1; *Peak GPU MB* is the peak CUDA memory during inference; *Host RAM GB* is peak host memory on the CPU edge proxy. CPU-edge latency and host memory were not measured for USAD or TranAD (–); GDN and Isolation Forest are not profiled here.

| Method                   | Params (W/H/S)        | Train s (W/H/S)   | GPU ms/win (W/H/S) | Peak GPU MB (W/H/S) | CPU-edge ms/win (W/H/S) | Host RAM GB (W/H/S) |
|--------------------------|-----------------------|-------------------|--------------------|---------------------|-------------------------|---------------------|
| LatAD (global density)   | 209k / 112k / 100k    | 3.2 / 16 / 3.5    | 0.7 / 13.0 / 13.0  | 19.9 / 19.4 / 19.7  | 0.6 / 14.6 / 14.5       | 3.5 / 3.7 / 3.9     |
| LatAD (regime-community) | 1.25M / 0.81M / 0.78M | 50 / 195 / 38     | 32.4 / 18.5 / 17.2 | 28.7 / 25.1 / 24.8  | 25.0 / 16.2 / 14.5      | 3.5 / 3.8 / 3.9     |
| TranAD                   | 1.27M / 0.30M / 0.23M | 231 / 898 / 163   | 2.0 / 2.0 / 1.9    | 62.7 / 42.3 / 38.9  | –                       | –                   |
| USAD                     | 32k / 16k / 14k       | 96 / 592 / 95     | 0.41 / 0.41 / 0.41 | 12.2 / 19.4 / 19.2  | –                       | –                   |
| AutoEncoder              | 97k / 48k / 42k       | 0.2 / <0.1 / <0.1 | 0.16 / 0.15 / 0.15 | 28.9 / 45.6 / 24.2  | 0.08 / 0.07 / 0.07      | –                   |

## Appendix C. Between-regime overlap (A8) screen

This appendix reports the measurements behind the A8 conclusion in §8. Cluster responsibility (how confidently the fitted mixture assigns each normal window to one regime) is not a valid overlap measure: a softmax posterior is crisp both when a window sits at a component and when it is far from all of them, so it cannot see a between-mode pocket. On a synthetic pair of regimes whose Bayes error runs from 30 percent to 2 percent the responsibility entropy stays flat, and it assigns a genuine between-mode window (observation-density percentile 0.000) responsibility 1.000. We therefore measure A8 in observation space, with four label-free quantities on the standardized window features: the *close-distinct pair mass* (the share of train-normal mass in a discovered regime that has a density-valley-separated neighbor within twice its own normal envelope, the A8 precondition), the *nearest-pair overlap coefficient* and density-valley ratio (heavy mode overlap itself), the labeled *anomaly-masking rate* beside its lag-matched regime-switch base rate (an anomaly of one regime absorbed by an adjacent one), and the *isolated-and-missed rate by head resolution* (the over-coverage failure of a low-resolution density head). A synthetic control calibrates them: a 30-percent-overlap regime pair (the positive control in Table C1). Table C1 reports the three benchmarks, the physically labeled Cranfield rig [61], SKAB, and the control; a broader screen (adding MetroPT [56], four SMD server-machine entities [31], and the Paderborn bearing testbed [62]) gives the same picture. The A8 precondition holds

on every multi-loop plant (close-distinct mass 0.99 to 1.00 in Table C1, 0.94 to 1.00 across the broader screen); the overlap, the masking excess over the base rate, and the over-coverage miss of the 80-component head are all far below the positive control.

**Table C1.** Between-regime overlap screen in observation space (label-free unless noted). *Close-distinct mass* is the share of train-normal mass in a discovered regime with a valley-separated neighbor within twice its own envelope, with the smallest distinct-pair Mahalanobis distance  $D$  in parentheses (the A8 precondition; higher mass, smaller  $D$  = more adjacency). *Overlap coefficient* is the nearest discovered-regime pair mean (max); heavy overlap would approach the 30-percent-Bayes-error control (0.61). *Masking rate / base* is the labeled anomaly-masking rate beside its lag-matched regime-switch base rate; an excess would indicate A8-masking. *Isolated-missed* is the share of isolated anomalies scored below the 99th train percentile by a wide two-component head versus the 80-component head. Synthetic positive controls calibrate each column.

| Dataset                        | Close-distinct mass (min $D$ ) | Overlap coeff., mean (max) | Masking rate / base | Isolated-missed, wide / 80-comp |
|--------------------------------|--------------------------------|----------------------------|---------------------|---------------------------------|
| WADI                           | 1.00 (2.8)                     | 0.09 (0.27)                | 0.010 / 0.100       | 0.80 / 0.13                     |
| HAI                            | 1.00 (2.4)                     | 0.09 (0.26)                | 0.000 / 0.035       | 0.01 / 0.03                     |
| SWaT                           | 0.99 (3.9)                     | 0.02 (0.05)                | 0.008 / 0.117       | 0.06 / 0.005                    |
| Cranfield (physical setpoints) | 1.00 (2.2)                     | 0.016 (0.030)              | 0.000 / 0.005       | 0.28 / 0.000                    |
| SKAB                           | 0.00 (4.3)                     | —                          | 0.002 / 0.010       | 0.017 / 0.000                   |
| positive control               | 1.00 (6.3)                     | 0.61 (30% overlap)         | 0.123 / 0.111       | 0.997 / 0.006                   |

**SKAB.** SKAB [63], a 400-window rotor rig, has no close-distinct regime pairs (0 of 12; it is a single-loop rig), so it does not exhibit the A8 precondition. A8 is specified for completeness but not demonstrated on any dataset in this study.

## Appendix D. Implementation and reproducibility

LatAD and the local feature baselines share the same windowing and features; the deep sequence baselines (USAD, TranAD, GDN) use their native inputs and are compared on the same labels and evaluation grid (§5.5). Raw channels are standardized per channel on train-normal statistics and, on WADI and SWaT, clipped to  $\pm 10$  standard deviations (HAI is not clipped; §5.1); the stream is then windowed at  $W = 60$  with stride 30, and each window is summarized by six per-channel statistics (mean, standard deviation, minimum, maximum, last-first, and range), concatenated in stat-blocked order, giving a feature dimension of 6C (732 for WADI, 354 for HAI, 306 for SWaT). A window is labeled anomalous when more than 5% of its raw timesteps carry an attack label. All multi-seed detectors use seeds 0–4.

### D.1 VaDE and detector hyperparameters

Table D1 lists the configuration of the global VaDE and its scoring heads. The mixture-component count  $K$  is a fixed per-dataset hyperparameter set a priori (20/40/40), not selected on any split; the BIC-optimal component count on train-normal windows (22/24/25) is reported separately as a characterization of regime multiplicity and is not used to tune the detector. The density head is a diagonal Gaussian mixture with  $M = 80$  components fit on the train-normal latent means; the residual head reduces the reconstruction residual to 30 principal components and whitens it with a per-regime Ledoit-Wolf covariance, and is auto-gated on by a train-normal held-out generalization check (active on HAI and SWaT, off on WADI).

**Table D1.** VaDE and detector hyperparameters. Values shared across datasets are given once; per-dataset values are listed as WADI / HAI / SWaT.

| Hyperparameter                                | Value                                                              |
|-----------------------------------------------|--------------------------------------------------------------------|
| Encoder / decoder multilayer perceptron (MLP) | [6C, 128, 64] → latent; decoder mirror, ReLU between hidden layers |
| Mixture components $K$ (fixed a priori)       | 20 / 40 / 40                                                       |
| Latent dimension                              | 10 / 16 / 16                                                       |
| Pretrain (plain-VAE) epochs                   | 30                                                                 |
| Joint epochs / warmup                         | 40 / 8                                                             |
| Batch size                                    | 256                                                                |
| Optimizer / learning rate                     | Adam / $1 \times 10^{-3}$                                          |
| Mixture-parameter lr ratio                    | 0.1                                                                |
| LR schedule                                   | StepLR (step 20, $\gamma$ 0.5)                                     |
| Component variance floor                      | $\log(0.05)$                                                       |
| Covariance-regularization weight              | $1 \times 10^{-3}$                                                 |
| Density head components $M$                   | 80 (diagonal GMM)                                                  |
| Residual PCA dimension                        | 30 (per-regime Ledoit-Wolf whitening)                              |
| Seeds                                         | 0, 1, 2, 3, 4                                                      |

## D.2 Regime communities, aggregation, and the reported detector

Communities are built from the train-normal window-mean correlation graph: near-constant channels (train standard deviation  $\leq 10^{-6}$ ) are dropped, a Pearson correlation matrix is formed on the standardized window means, the distance is  $1 - |q|$ , and average-linkage hierarchical clustering is cut into every nested dendrogram subtree of size 3 to 25 channels (giving  $S = 44$  communities on WADI, 28 on HAI, 24 on SWaT). Each community's cohesion is its mean pairwise  $|q|$  and its fusion weight is cohesion  $\times \sqrt{\text{size}}$ . Each community trains a smaller VaDE ( $K = \min(20, \max(6, |G|))$ , latent  $\min(8, \max(3, |G|/2))$ , 15 epochs, density head  $\min(50, \max(12, \lfloor n_{\text{fit}}/12 \rfloor))$ ). For every model the first 80% of train-normal windows (chronological) is the fit slice and the last 20% is a held-out calibration slice used as the upper-tail reference; no test data enters fitting or calibration. Per-community upper-tail p-values are combined by cohesion-weighted Higher Criticism, and the reported detector, **LatAD (regime-community)**, is the standardized sum of that cohesion-weighted Higher-Criticism score and the whole-plant (global VaDE) upper-tail score, each standardized against its train-normal reference. The double-hard subset (Table A3) is constructed by a linear principal-component filter (Hotelling  $T^2$  and squared prediction error at their 99th train-normal percentiles, 95% retained variance), and the drift-aware typing (§7) decomposes each community's surprise against a causal 24 h rolling median. Baseline settings are in Table D2.

**Table D2.** Baseline detector settings. Isolation Forest, the AutoEncoder, USAD, and TranAD use five seeds; the linear predictor, boosted linear predictor, and GDN are single-seed.

| Baseline          | Settings                                                                                                                                    |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Isolation Forest  | 200 estimators, default max-samples and contamination; score = $-\text{score\_samples}$                                                     |
| AutoEncoder       | 6C-64-16-64-6C, ReLU; Adam, lr $1 \times 10^{-3}$ , 40 epochs; score = summed squared residual                                              |
| LinRes (linear)   | ordinary least squares, leave-one-channel-out; discrete channels ( $\leq 6$ states) soft one-hot; score = mean per-channel squared residual |
| Boosted LinRes    | histogram gradient boosting (150 iterations, learning rate 0.1), same leave-one-channel-out structure                                       |
| USAD              | public TranAD evaluation harness [37], 30 epochs                                                                                            |
| TranAD            | same harness, 5 epochs                                                                                                                      |
| GDN               | same harness, window length 5, 5 epochs, harness-default top-k and embedding dimension; single seed, AUROC only                             |
| trivial $\max u $ | maximum absolute standardized window mean over the level (mean) statistic block                                                             |

## Appendix E. Rare-regime safety of the nearest-component score (A5)

The nearest-component likelihood (§4.3) is designed to avoid false alarms on rare-but-valid regimes: the  $\pi$ -weighted mixture negative log-likelihood penalizes a window drawn from a low-prior component, whereas the nearest-component score, which drops the  $\pi$  weight, judges a window only against its closest regime. The three benchmarks are effectively single-mode, so this property cannot be exercised on them; we test it on the 3W oil-well telemetry dataset [64], which has genuinely imbalanced regime occupancy (its rarest discovered regime holds under 0.1 percent of the train-normal mass). At a false-alarm rate matched between the two scores, the mixture false-alarms on rare regimes while the nearest-component score does not: across three seeds the number of test-normal false positives falling in a rare (under 5 percent occupancy) regime drops from 17, 17, 23 under the mixture to 11, 9, 16 under nearest-component (Table E1), and, at these genuinely rare (under 5 percent occupancy) regimes, the empirically observed invariant that nearest-component does not add a rare-regime false alarm relative to the mixture holds on every seed. Every false alarm removed sits in a low-occupancy regime (median occupancy 0.002 to 0.024, against 0.07 to 0.14 for all test-normal windows). Detection at the matched operating point is unchanged (difficult-subset true-positive rate at 5 percent false-alarm  $+0.007 \pm 0.068$ ; AUROC  $+0.003 \pm 0.020$ , three seeds), so the property is defensive: it removes rare-regime false alarms at no detection cost, the behavior the score is designed for and one the single-mode benchmarks cannot show.

**Table E1.** Rare-regime false-alarm safety on 3W (imbalanced regime occupancy). At a false-alarm rate matched between the two scores, test-normal false positives falling in a rare (under 5 percent occupancy) regime,  $\pi$ -weighted mixture versus nearest-component, three seeds, with the difficult-subset true-positive-rate change at 5 percent false-alarm. Nearest-component removes rare-regime false alarms (invariant: it never adds one) at no detection cost.

| Seed | Rare-regime FP (mixture) | Rare-regime FP (nearest) | Diff.-subset TPR@5% change |
|------|--------------------------|--------------------------|----------------------------|
| 0    | 17                       | 11                       | +0.064                     |
| 1    | 17                       | 9                        | +0.046                     |
| 2    | 23                       | 16                       | -0.089                     |
| mean | 19                       | 12                       | $+0.007 \pm 0.068$         |

To confirm the effect is not an artefact of one split or one comparison, we sweep the rare-regime occupancy threshold (2, 5, and 10 percent) and compare the nearest-component score against three imbalance-aware alternatives: a uniform-prior mixture (a flat  $1/K$  prior), a tempered mixture ( $\pi^{0.5}$ ), and a  $k$ -nearest-neighbour latent density. Reported as a rare-regime false-positive rate (rare-regime false positives divided by the rare-regime normal count) at a matched overall false-positive rate, the nearest-component score is at or below every alternative at every threshold on 3W (Table E2): at the 5 percent threshold it reaches 0.148 against 0.210 for the  $\pi$ -weighted mixture, with the prior-flattening alternatives in between (uniform-prior 0.204, tempered 0.205) and the non-parametric kNN worst (0.216). The gain therefore comes specifically from dropping the  $-\log \pi_k$  rare-regime penalty, not from generic imbalance-awareness. Difficult-subset detection at the matched operating point is unchanged (nearest 0.299 against mixture 0.318 true-positive rate, within seed variation). At the genuinely rare thresholds ( $\leq 5$  percent) the nearest score never adds a rare-regime false positive (0 of 30 seed $\times$ operating-point cases per threshold); at a looser 10 percent threshold, which dilutes “rare” to moderately-common regimes, it protects the rarest regimes, where the prior penalty is largest, and can redistribute a few flags onto moderate tails (1 of 30).

**Table E2.** Rare-regime false-positive rate on 3W across rare-occupancy thresholds (2, 5, and 10 percent) and scoring rules, at a matched overall false-positive rate (5 percent), five-seed mean±standard deviation. Lower is better. Scores:  $\pi$ -weighted mixture (naïve), nearest-component (A5), uniform-prior mixture, tempered ( $\pi^{0.5}$ ) mixture, and  $k$ -nearest-neighbour latent density. Nearest-component is at or below every alternative at every threshold; the prior-flattening alternatives fall between the mixture and nearest, isolating the  $-\log \pi$  penalty as the mechanism.

| Rare-occupancy threshold | $\pi$ -mixture | nearest (A5) | uniform-prior | tempered    | kNN         |
|--------------------------|----------------|--------------|---------------|-------------|-------------|
| 2 percent                | 0.453±0.121    | 0.316±0.129  | 0.431±0.105   | 0.436±0.110 | 0.518±0.110 |
| 5 percent                | 0.210±0.049    | 0.148±0.042  | 0.204±0.052   | 0.205±0.051 | 0.216±0.047 |
| 10 percent               | 0.111±0.025    | 0.092±0.027  | 0.108±0.024   | 0.109±0.024 | 0.116±0.027 |

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